

MTH601 Short Notes Mid term

Lec 1 to22

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Mth601

Operations Research

Segment I: Introduction

Lectures 1-3

Background

Technology - the application of science to the everyday Physical World - has, on the whole, changed our lives greatly for the better, particularly since the time of Industrial Revolution. Manufacturing industry, which supplies us with and at the same time creates the wealth, which enables other services to exist, owes its existence to Science and Technology.

Some manufactured items, aircraft for example, are both complex and expensive, and there are great difficulties, first of all in organizing their manufacture and then, once made, in using them efficiently. Problems of this latter kind, often called systems problems, were studied rigorously for the first time by scientists, not in industry but in the management of operations in the 1939 - 45 war.

A number of teams of eminent scientists were employed by the British Government to apply their expertise to management and operational problems rather than technical problems. One such team (the Radar Operational Research team - which gave OR its name) was responsible for implementing the installation and operation of radar after the technical development work was complete.

Later on another team examined the relative ineffectiveness of the Allied Forces at destroying the German U-boats, which were sinking the food convoy ships supplying Britain. The OR team played an important part in reducing shipping losses and in the ultimate defeat of the U-boats. With no precedent in the application of Science to the management function, the scientists were still able to use their scientific approach-collecting information and developing hypothesis in order to come up with practical plans for the improvement of these wartime operations.

OR

The art of winning war without actually fighting it

After the end of the war II ,

The same approach was used in industrial and commercial field.

First developed in Britain and simultaneously in USA followed by Europe and rest of the world.

But not yet in Pakistan! Why?

After the end of the war II, the same approach was used with great success in the industrial and commercial field, developing most rapidly in Britain and the USA, followed by Europe and the rest of the world. This has provided the basis for the evolution of Operational Research (OR) as a separate and independent subject of Science. It is not based on any single academic engineering, social science, economics, statistics and computing but is none of these. Then what is OR?

Definition of OR,

DEFINING OPERATIONS RESEARCH:

OR has been defined in various ways and it is perhaps too early to define it in some authoritative way. However given below are a few opinions about the definition of OR which have been changed along-with the development of the subject.

In 1946 Morse & Kimbel has defined as;

"OR is a scientific method of providing executive departments with a quantitative basis for decision regarding the operations under their control"

In 1948 Blackett defined as;

"OR is a scientific method of providing executives with any analytical and objective basis for decisions"

Another definition is due to Morse who defined in 1948 as;

"The term OR, has here-to fore been used to connote various attempts to study operations of war by scientific methods. From a more general point of view, OR can be considered to be an attempt to study those operations of modern society which involved organizations of men or men and machines".

Later on in 1957, Churchmen Ackoff and Arnoff defined;

"OR is the application of scientific methods, techniques and tools to problems involving the operations of systems so as to provide those in control of the operations with optimum solutions to the problem".

In 1958 Saaty defined OR as;

"The art of giving bad answer to problems to which, otherwise, worse answers are given".

The Operational Research Society of U.K. defines OR as:

"Operational Research is the application of the methods of science to complex problems arising in the direction and management of large systems of men, machines, materials and money in industry, business, government and defense. The distinctive approach is to develop a scientific model of the system, incorporating measurements of factors, such as chance and risk, with which to compare the outcome of alternative decisions, strategies and controls. The purpose is to help management determine its policies and actions scientifically."

In the USA, where it is called Operations Research, the OR Society of America says more briefly;

"OR is concerned with scientifically deciding how to best design and operate man-machine systems, usually under conditions requiring the allocation of scarce resources".

An even briefer definition might be "Science applied to management", but however, it might be defined, there is no doubt that OR provides the numerate scientist - of whatever discipline - with an opportunity to apply the skills of science in the field of Management.

Before proceeding further let us define for the sake of clarity some fundamental terms.

MANAGEMENT, MANAGEMENT SCIENCE AND OR:

MANAGEMENT may be equated with decision-making and control. Government ministers manage the economy industrialists make decision within their companies and individual make personal decisions.

MANAGEMENT SCIENCE is the study of problems as abstractions and the application of the resulting theory to practical situations. Its two fundamental disciplines are behavioral science and science and quantitative methods.

OPERATIONS RESEARCH (OR) is the application of quantitative methods to decision making. It formulates problems incisively and assesses the possible consequence of alternative course of action, so that informed and effective decisions can be taken.

OR APPROACH TO PROBLEM SOLVING:

OR encompasses a logical systematic approach to problem solving. This approach to problem solving as shown in fig. 1 follows a generally recognized ordered set of steps: (1) observation, (2) definition of the problems, (3) model construction, (4) model solution, and (5) implementation of solution results.

OR PROCESS

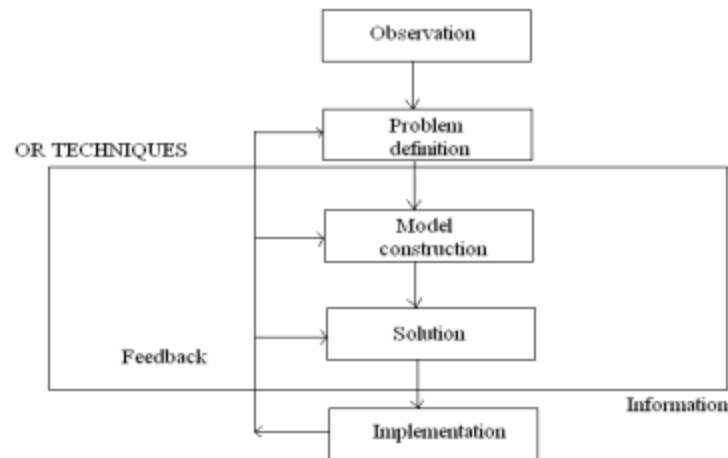


FIGURE 1

Observation

The first step in a problem solving exercises in OR is the identification of a problem that exist in the system. This requires that the system be continuously and closely observed so that problems can be identified as soon as they occur.

Definition of the Problem

Once it has determined that a problem exists, it must be clearly and concisely defined. The problem definition includes the limits of the problems and the degree to which it pervades other organs of the system. A requirement of problem definition is that the goals (or objective) must be clearly defined which helps to focus attention on what the problem is.

Model Construction

An OR model is an abstract representation of an existing problem situation. It can be in the form of a graph or chart, but mostly, an OR model consists of a set of mathematical relationship. In OR terminology, these are called objective function and constraints.

Model Solution

Once models are constructed, they are solved using the OR techniques, presented in the next section. Actually it is difficult to separate model construction and solution in most cases, since OR technique usually applies to a specific type of model. Thus, the model type and solution method are both part of the OR technique.

Implementation of Results

The results of an OR technique are information which helps in making a decision. The beauty of OR process lies in obtaining, the results which are implement able or we call it a feasible whole exercise will go waste.

OR is an On-going Process

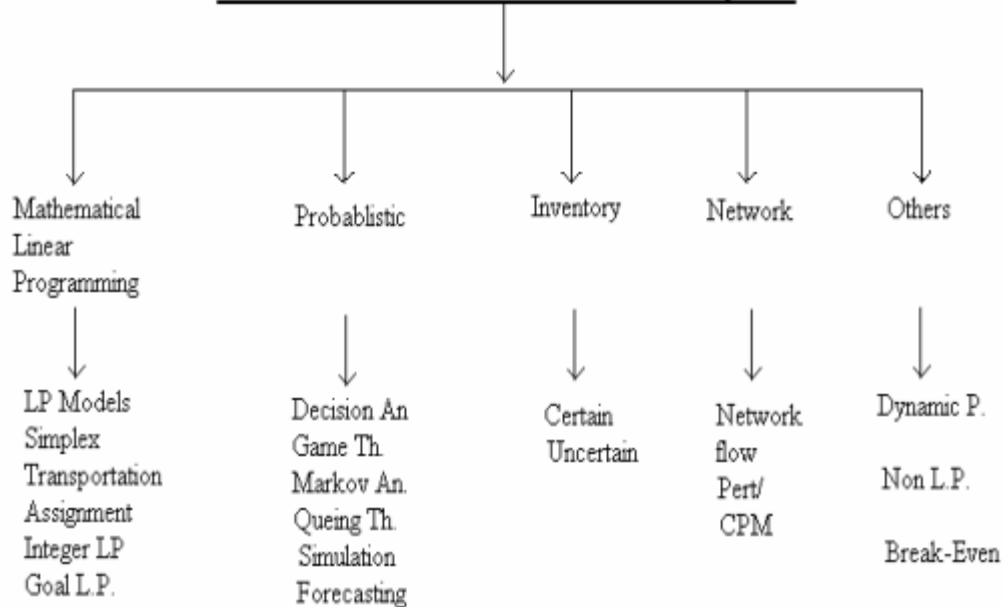
Once the five steps described above are completed, it does not necessarily mean that OR process is completed. The model results and the decisions based on the results provide feedback to the original model. The original OR model can be modified to test different conditions and decisions that might occur in the future. The results may indicate that a different problem exists that had not been thought of previously, thus the original model is altered or reconstructed. As such, the OR process is continuous rather than simply consisting of one solution to one problem.

Operations Research Techniques:

Two of the five steps of OR process, model construction and solution, encompass the actual use of OR techniques. These techniques can be loosely classified into five categories.

- 1) Linear mathematical programming technique consist of first, identifying problem as being solvable by linear programming; second formulation of unturned problem and then finding the solution by using established mathematical techniques. It derives its name from the fact that the functional relationship in the mathematical model are linear and the solution techniques consists of a predetermined mathematical steps i.e. program.
- 2) Probabilistic techniques covers those problem in which all parameters are not known with certainty. The solution results are assumed to be known with uncertainty, with probability that other solution might exist.
- 3) Inventory techniques are specifically designed for the analysis of inventory problem frequently encountered by the business firms. This particular business function is singled out for attention, since it typically represents a significant area of cost for almost every business. This category is divided into probabilistic and deterministic techniques.

OPERATIONS RESEARCH TECHNIQUES



4) Network techniques consist of models that are represented by diagrams rather than strictly mathematical relationship i.e. pictorial representation of the system under consideration. These models can represent either probabilistic or deterministic systems.

5) Other techniques consist of all the remaining techniques, which do not come under the four heads mentioned above. For example, Dynamic programming employs a different modeling and solution logic than linear programming. In non-linear programming either the objective function or the constraints or both can be non-linear functions, which would require altogether different solution technique.

USES OF OPERATIONS RESEARCH

In its recent years of organized development, OR has entered successfully many different areas of research for military government and industry in many countries of the world. The basic problem in most of the developing countries in Asia and Africa is to remove poverty and improve the standard of living of a common man as quickly as possible. So there is a great scope for economists, statisticians, administrators, politicians and the technicians working in a team to solve this problem by an OR approach. The possible application sectors, in Pakistan, are as under:-

1) Macro Economic Planning:

OR can be employed for Macro-Economic Planning of the country:

- a) Input / Output Analysis: by using LP models. This input/output analysis can be of any duration [i.e. of short term (up to say 10 years)-Five Year Plan; and of long term (10-30 years)].
- b) Investment Planning: OR can be employed in the Investment Planning of the country where investment plans for the next five or ten years are prepared. Mixed Integer Programming and Linear Programming techniques can be used.
- c) Choice of Projects: OR can help the people in the planning in choosing the optimal project. This sort of choice would need Integer Programming and Quadratic Assignment techniques.
- d) OR can also be used in Simulation Modeling of the Economy of the country.

2) **Sectoral Planning:**

OR can also be employed in a particular sector of the Economy, e.g. in agriculture, in finance, in industry, in marketing, in production, in management etc.

- a) Scheduling all operations within a sector can be done by using OR e.g. production scheduling + Distribution planning + marketing + Personnel management + maintenance +
- b) Schedule of some operations within a sector can be done by employing OR e.g. Inventory planning in agriculture or distribution of fertilizer etc.

3) **Micro Economic Planning:**

This sort of activity involves for example:

Planning the operations of a Company.

Improving the layout of a workshop in a company.

Finding size of a hospital in an area etc.

There is a great potential for utilizing OR in this area of planning in our country.

POTENTIAL AREAS OF APPLICATIONS

As mentioned earlier OR can be applied in every field of life. Here are a few of the many fields where OR has potential application. This list is by no means comprehensive or exhaustive but definitely will provide an idea of the power of OR as a separate discipline.

Operations Research in the Public Sector

Federal, Provincial and Local Government

- Development of Country Structure Plans
- Manpower Planning and Career Development in Govt. Departments
- Organization of Long-Term planning groups at the National Level
- Corporate Planning in Local Government
- Allocation of Government Houses
- Estimation of Future Requirement of School/College Building
- Placing of Fire Brigade in a City
- Measuring the Effectiveness of Police
- Timetabling in Schools and Colleges for Efficient use of Space

Health

- Management policies for 120-bed nursing units
- Optimum size of general hospitals
- Appointment systems for hospital outpatients
- Stock control for regional and area health units
- National and area planning of health services
- Manpower planning for nurses, radiographers, etc.
- Commissioning of a new general hospital
- Simulation of pathology laboratories
- Organizing an ambulance service
- Care provided by community nurses

Defense

- Arms control and disarmament studies
- Communications network development
- Logistic support in operations
- Field experimentation
- War games and other models of battle
- Equipment procurement
- Reinforcement and redeployment problems

Operations Research in Industry & Commerce

Finance and Investment

- Developing the five-year plan for a food manufacturer
- Development of the pipeline
- Computer based financial planning
- Portfolio selection
- Structure for the assets of a bank
- Evaluating investment in a new plant
- Corporate planning in the chemical industry
- Financing expansion of a small firm

Production

- Production scheduling in a steel works
- Meeting peak demands for electricity
- Minimization of costs of power station maintenance
- Scheduling newsprint deliveries
- Stock levels of steel plate
- Meeting seasonal demands for products
- Blending scrap metals
- Stock policy for a paint manufacturer
- Allowing for yarn breaks in spinning
- Meeting customer requirements for carpets
- Planning a quarry's output
- Optimum layout for belt coal transport in a colliery

Marketing

- Launching a new product
- Advertising effectiveness and cost
- Planning sales territories
- Measurement of consumer loyalty
- Buyer-seller behavior
- Advertising research and media scheduling

- Most profitable retail brand mix
- Developing customer service policies
- Pricing policies for confectionery

Personnel

- Personnel shift planning
- Manpower planning
- Manpower for an assembly line
- Effects of flexible working hours

Distribution

- Distribution of Products.
- Returnable bottles: how many?
- Refinery crude tank capacity
- Depot location of pharmaceutical products
- Trucking policy for dairy products
- Distribution of newspapers to newsagents

OR in Transport

Rail

- Rail freight management
- Required fleet size of locomotives and rolling stock
- Forecasting passenger traffic
- Planning reconstruction of main-line termini
- Introduction of freightliners

Road

- Designing urban road networks
- Forecasts of car ownership
- Implementation of bus lanes

- Re-routing bus services
- Purchasing and maintenance of buses
- Introduction of flat-fare buses
- Bus services in rural areas
- Preparation of crew rosters

Air

Planning the introduction of Boeing 737/Airbus 300

Allocation of aircraft and crew to routes

Location of Islamabad Airport

Karachi-Lahore - Islamabad - Peshawar: aircraft requirements

Sea

- Potential traffic for new container services
- Shipbuilding requirement in the 1990's
- Optimum ship size for given routes
- Construction and management of a container terminal

EXAMPLE

Before proceeding further let us take an example, which will help to understand the scope of application in various activities. Given below are some of the major activities which

OPERATIONS:

1) Oil production from fields

2) Transportation of Crude

- from fields to refineries
- from fields to export ports (Jetties)
- from import ports (Jetties) to refineries

3) Storage of Crude

- on fields

- at Ports
- at refineries

4) Refinery Scheduling

- Operation of CDU's
- Operation of Blending Units

5) Storage of Distilled Blended Products

- at refineries
- at Jetties
- at distribution points

6) Transportation of Products

- from Jetties to refineries
- from one refinery to another for another processing
- from refinery to bulk distribution pts.
- from bulk distribution points to final consumers.

At all the stages from oil production from fields to its transportation to the final consumer OR has been employed in the developed countries of the world. Applying on macro level is not an easy job. This would require true and factual data computing power and trained professionals, and perhaps at this stage we may face some problem due to limited resources in term of manpower, money and machines, but it does not mean that we should not make a beginning.

REQUIREMENTS FOR APPLYING SUCCESSFUL OR

In order to apply OR successfully in achieving certain objectives the following are some of the prerequisites:

- 1) Sufficient number of qualified OR personnel should be available who should know the art of selling OR. These professional should be ready to work in close cooperation with staff at various levels of the organization because they have to dig out the problem from the grass roots.
- 2) Normally, there is a tendency of management to give false information, inorder to create a better image, and sometime there may be a lack of adequate communication between OR practitioner and management. To meet these circumstances there should be a healthy interaction with the statistical machinery so as to enhance the quality and availability of data.

- 3) Satisfactory power of computer machinery along with associated support should be available so as to analyze the data and forecast correctly. For this management should be aware of modern techniques and tools of management.
- 4) Last but not the least management should be ready to accept the change and be enthusiastic to know the correct alternatives.

CONCLUSION

In the proceeding paras we traced briefly the history of OR; defined OR, and its process; elaborated various techniques of OR; identified various areas of application and requirements for successfully applying OR. Before concluding it will be most appropriate to quote Stafford Beer, the famous OR Scientists.

"We call that work operational (with a large O) because it is based in the world of genuine activity, the places where things actually happen. All good sciences, as distinguished from all mysticism is found in empiricism. It involves actual observation, actual measurement and actual experiment."

We call our work Research (with a capital R) because we deal with problems to which no one knows the answer.

"Doing that thing is called RESEARCH!"

Finally I would say,

"We need OR teams of outstanding abilities, working on problems of decision & control at the National Level. Because these problems are usually discussed in economic terms, they are currently assumed to be purely economic problems."

BUT THEY ARE NOT

"Interdisciplinary Scientific teams are needed to evaluate issues subject to conflicting criteria."

REFERENCES

Bellman, Richard E. Dynamic Programming. Princeton University. Press

Gale, David, The Theory of Linear Economic Models. McGraw-Hill

Hadley, G., Linear Programming. Addison Wesley

Ackoff, R. L., and P. Rivett, A Manager's Guide to Operations Research, Wiley

Dantzig, George B., Linear Programming and Extensions, Princeton University Press

Hadley, G., Nonlinear and Dynamic Programming, Addison Wesley

Ackoff, Russell L., and Maurice W. Sasieni, Fundamentals of Operations Research, Wiley

Beale, E. M. L., Ed. Applications of Mathematical Programming Techniques. English University. Press

Rivett, P., Principles of Model Building, Wiley

Wagner, Harvey M. Principles of Operations Research, Prentice-Hall

Taha, Hamdy A., Operations Research: An Introduction, Macmillan

Ignizio, J.P., Linear Programming in Single and Multiple Objective System. Prentice-Hall

Segment II: PERT / CPM

Lectures 4 -10

INTRODUCTION

PERT is an acronym for "Programme Evaluation and Review Technique". This was created as a means to plan and accelerate the development of the Polaris Ballistic Missile. In USA the defence department developed a nuclear missile to be launched from beneath the ocean's surface by a mobile submarine, which would be an effective deterrent against aggression by an enemy. This paved way to plan how to design, develop and plan the different stages in the production of a missile and how quickly this task could be completed. A planning and scheduling technique named PERT gave the answer to these questions.

In any new venture, uncertainties are bound to creep in. PERT incorporated these uncertainties into a model, which provides a reasonable answer to these uncertainties. There are certain statistical aspects scheduling large projects consisting of numerous activities whose completion times are uncertain and are independent of one another. PERT is an event-oriented technique. By 'event' we mean reaching a certain stage of completion of the project.

Another technique, Critical Path Method, abbreviated as CPM, has emerged simultaneously. It is also a network technique but it is concerned with obtaining the trade-off, between cost and completion date for large projects. In any project consisting of several activities each activity can be completed in a normal duration with normal cost. If we employ more persons or skilled people or given overtime to the workers, the activity could be completed in a reduced duration known as crash duration. But this involves an increased cost in the form of additional resources. With CPM the amount needed to complete the various activities is assumed to be known with

certainty. So, the direct costs for the activities increase and hence the cost of the project also increases. By reducing the activity duration of some or all possible completed ahead of the schedule. This will naturally reduce the overhead cost for the entire project. On one hand the direct expenses increase, if we shorten the activity duration, but, the indirect expenses for the project are reduced. We have to strike a balance or an optimum time schedule, or a least cost schedule is to be obtained. This is the purpose of the Critical Path Method. Thus CPM is not concerned with uncertain job times as in PERT. PERT is useful in research and developmental projects, whereas CPM is mostly used in construction projects, or in situations already handled, so that the details like the normal completion time, crash duration and cost of crashing are already known.

The following are the suggested applications when PERT or CPM is found useful.

- The construction of a building or of a highway.
- Planning and launching a new product.
- Scheduling maintenance for a project.
- The manufacture and assembly of a large machine tool.
- To conduct a music or drama festival.
- Preparation of budget for a company.

CONCEPT OF NETWORK

The first step in the application of CPM / PERT is to develop a network representation of the project plan.

A 'network' is defined as a graphic representation with a flow of some type in its branches. It represents nodes and branches. Below in table 1, we represent different systems satisfying the definition of network in the physical world.

Table 5.1

Physical situation	Nodes	Branches	Flow
Highway systems	Intersections	Roads	Vehicles
Communication systems	Switching points	Wires	Messages
Fluid supply systems	Pumping stations	Pipes	Fluid
Production systems	Work centres	Handling routes	Jobs
Project Management	Decision points	Activities	Time
Airway systems	Airports	Airlines	Aircraft

A node is the intersection of the two branch lines. It is denoted by a circle. Each branch represents an activity. Each node represents an event, which is a specific definable accomplishment recognizable at a particular instant of time. The arrowheads indicate the sequence in which events must be achieved. Thus an event is the completion of all the activities leading into that node and this event must precede the initiation of the activity leading out of the node.

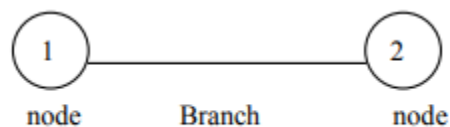


Fig. 1

An arrow diagram represents a project graph. An arrow connecting two nodes, representing two events, represents each activity. The head of the arrow identifies the start of the activity.

Let us take an example and illustrate the construction of arrow diagram for a project.

Example 1: A company is interested in preparing a budget. The details of the activities and the departments involved are given in the Table 2.

Table 2

The project of preparation of a Production budget.

Job Identification	Alternate	Job Description	Department
A	1-2	Forecasting sales	Sales
B	2-4	Pricing Sales	Sales
C	2-3	Preparing production Schedule	Engineering
D	3-4	Costing the production	Costing
E	4-5	Preparation of budget	President

The project of budgeting can be displayed in a network or project graph by an arrow diagram. Jobs are shown as arrows leading from one node to the other as in Figure 2.

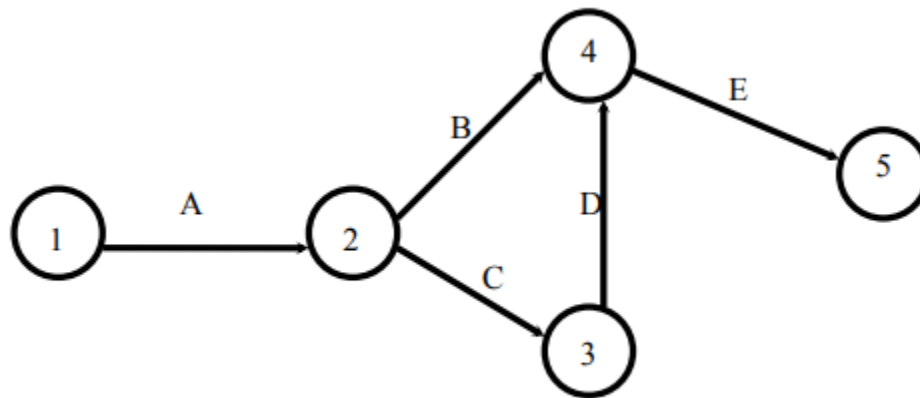


Fig. 2

From the arrow diagram 2, we infer that activity A is the first job. Jobs B and C start only after A is over. A is called the predecessor of B and C and B and C the successors of A.

RULES FOR CONSTRUCTION OF NETWORK

- (a) Each activity is represented by one and only one arrow. This means that no single activity can be represented twice in a network.
- (b) No two activities can be identified by the same end events. This means that there should not be loops in the network.
- (c) Time follows from left to right. All the arrows point in one direction. Arrows pointing in opposite direction must be avoided.
- (d) Arrows should not cross each other.
- (e) Every node must have at least one activity preceding it and at least one activity following it, except for the nodes at the very beginning and at the very end of the network.

DUMMY ACTIVITIES

There is a need for dummy activities when the project contains groups of two or more jobs which have common predecessors. The time taken for the dummy activities is zero.

Suppose we have the following project of jobs with their immediate predecessors.

Jobs	Immediate
------	-----------

	predecessors
A	-
B	-
C	-
D	A, B
E	B, C

Activity B is the common immediate predecessor of both D and E. A is the immediate predecessor of D alone and C is the predecessor of E. Let B lead into two dummy jobs D1 and D2 and let D1 be an immediate predecessor of D and D2 of E as shown in Figure 3.

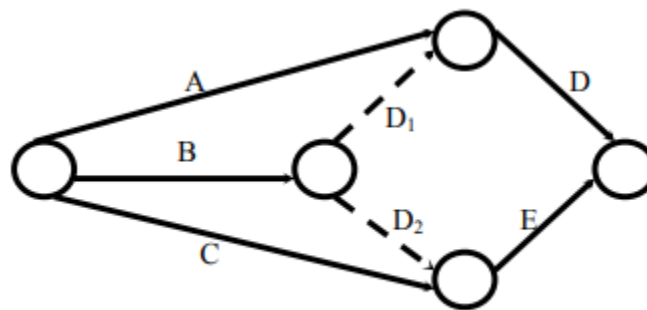


Fig. 3

Dummy arrow represents an activity with zero time duration. It is represented by a dotted line and is introduced in a network to clarify activity pattern under the following situations.

- (i) It is created to make activities with common starting and finishing events distinguishable.
- (ii) To identify and maintain the proper precedence relationship between activities those are not connected by events.

Consider an example where A and B are parallel (concurrent activities). C is dependent on A and D is dependent on both A and B. Such a situation can be represented easily by use of dummy activity.

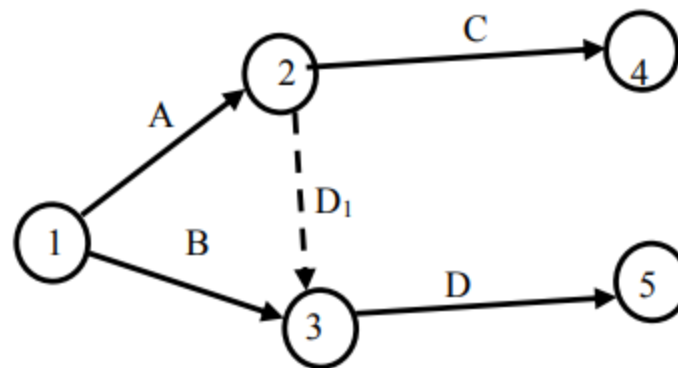


Fig. 4

Example 2: A project consists of the following activities whose precedence relationship is given below. Draw an arrow diagram to represent the project.

Activity	Followed by	Preceded by
A	B, C	-
B	D	C, A
C	E, B	A
D	F	B
E	G, F	C
F	H	E, D
G	H, I	E
H	I	G, F
I	-	G, H

Solution

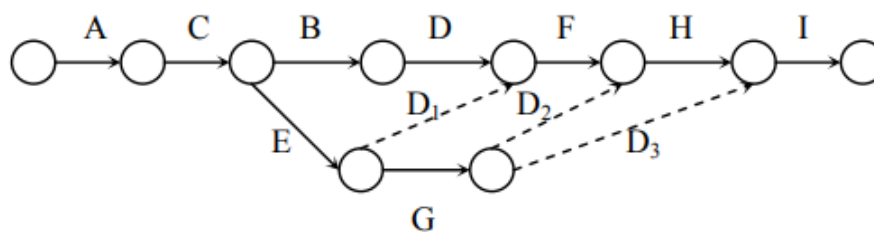


Fig. 5

Example: Consider the following project. Draw an arrow diagram to represent the project.

Activity	A	B	C	D	F	G	H	I
Precedence	-	-	A	A	B, C	F	B, C, D	H, G

Solution:

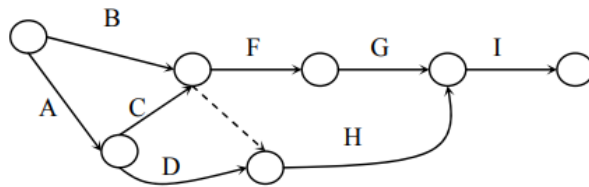


Fig. 6

Example Consider the project of building a house. The details of the project activities are tabulated below. Draw network.

Activity	A	B	C	D	E	F	G	H	I	J	K
Immediate Predecessor	-	A	-	B,C	C	G,H	D	B	F	G	E,I,J

Solution:

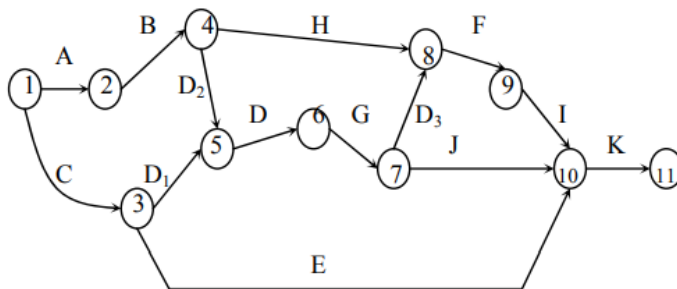


Fig. 7

Example A project has the following activities. The relationships among the activities are given below. Construct the network.

A is the first operation.
 B and C can be performed parallel and are immediate successors to A.
 D,E and F follow B.
 G follows E.
 H follows D, but it cannot start till E is complete.
 I and J succeed G.
 F and J precede K.
 H and I precede L.
 M succeeds L and K.
 The last operation N succeeds M and C.

Solution:

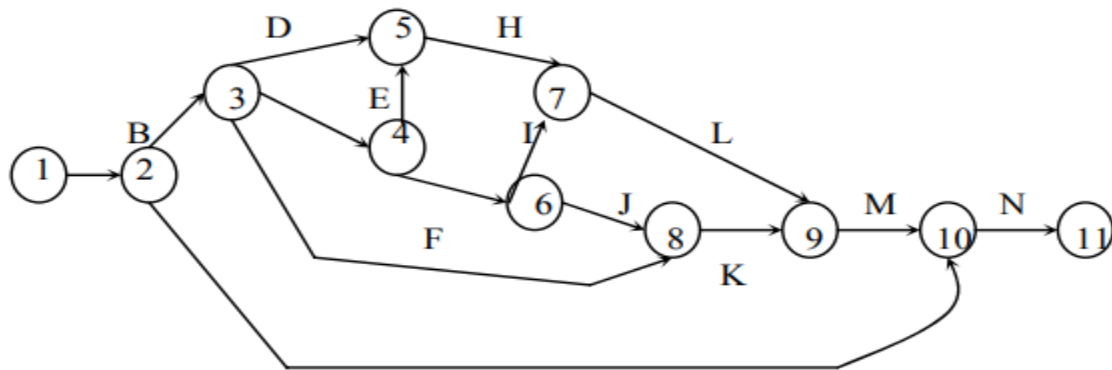


Fig. 8

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Activity	A	B	C	D	E	F	G	H	I	J	K	L
Predecessor	-	-	A	A	B	B	C,D	E	C,D	G,H	F	J,K

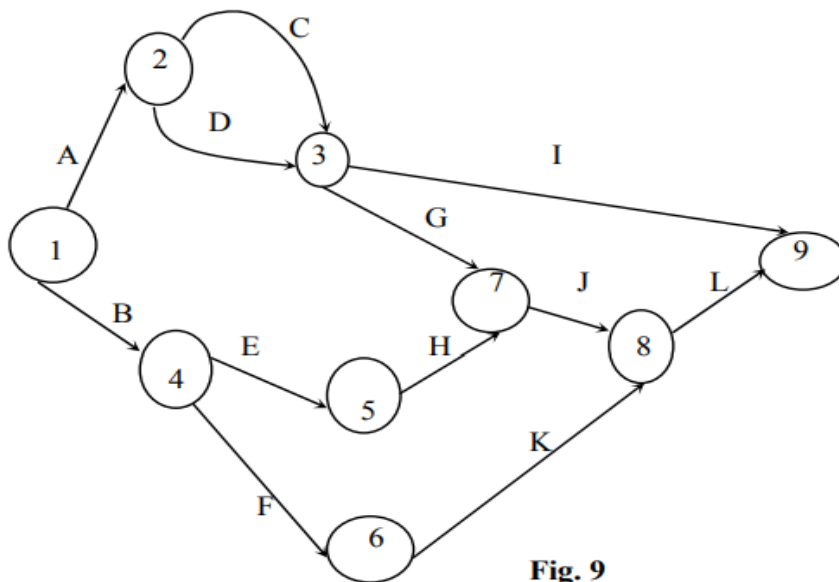
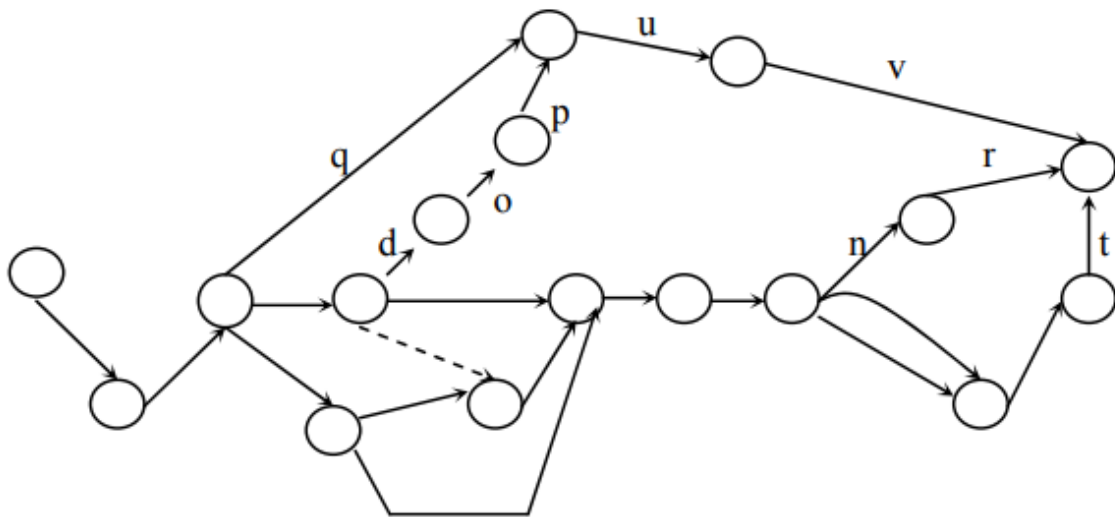


Fig. 9

Example Draw network from the following list of activities.

Job Name	Immediate predecessor	Job Name	Immediate predecessor
a	-	l	k
b	a	m	k
c	b	n	k
d	c	o	d
e	b	p	o
f	e	q	b
g	e	r	n
h	c	s	l, m
i	c, f	t	s
j	g, h, i	u	p, q
k	j	v	u

Example Draw network from the following list of activities.



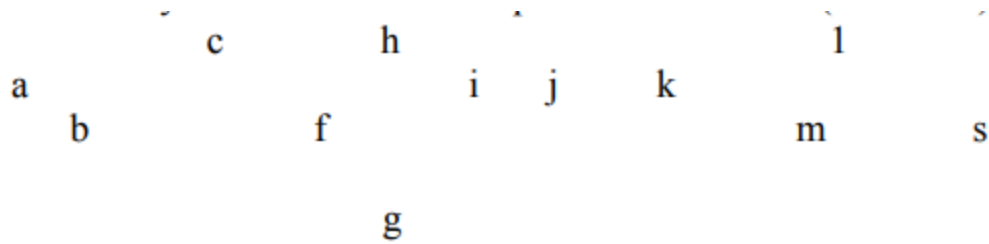


Fig. 10

Example For a project of 12 activities the details are given below. Draw PERT network.

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Dependence	-	-	-	B,C	A	C	E	E	D,F,H	E	I,J	G

Solution:

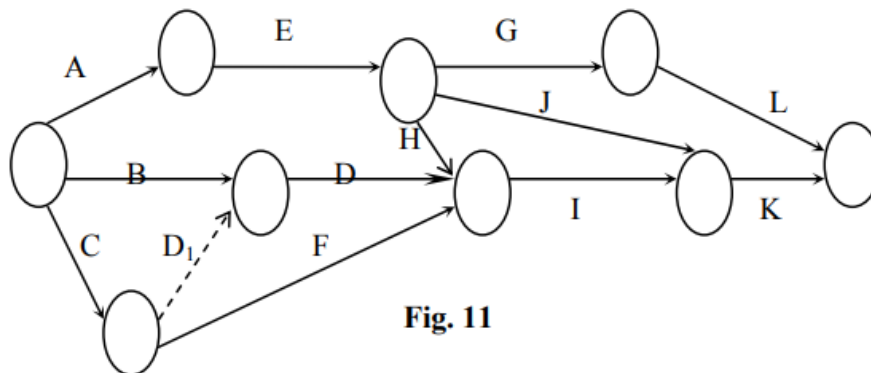


Fig. 11

Exercises

- (a) Explain in brief PERT, CPM and dummy activities with reference to Project Management.
- (b) The following information is known for a project. Draw the network and find the critical path. Capital letters denote activities and the numbers in bracket denote activities' time.

This must be completed	before	This can start
A(30)		C
B(7)		D
B		G
B		K
C(10)		D
C		G
D(14)		E
E(10)		F
F(7)		H
F		I
F		L

F	I
G(21)	L
H(7)	J(15)
I(12)	J
K(30)	L(15)

2. Draw network diagram for the following activities whose predecessors are given in the table below.

Job	A	B	C	D	E	F	G	H	I	J	K
Immediate Predecessors	-	A	B	C	B	E	D,F	E	H	G,I	J

TO FIND THE CRITICAL PATH

After listing all the activities with their precedence relationship we project these activities in a project graph represented by arrows or Activity On Node diagram (AON). Now we have to find the minimum time required for completion of the entire project. For this we must find the longest path with the sequence of connected activities through the network. This is called the critical path of the network and its length determines the time for completion of the project. The activities in the critical path are so critical that, if they are delayed, the project completion date cannot be met and the project finish time will have to be extended. We shall now see how to

identify the critical path, the critical activities and the duration of the project. The meaning of path and length of a path should first be made clear. Let us take following example.

Consider the following network shown in figure 12.

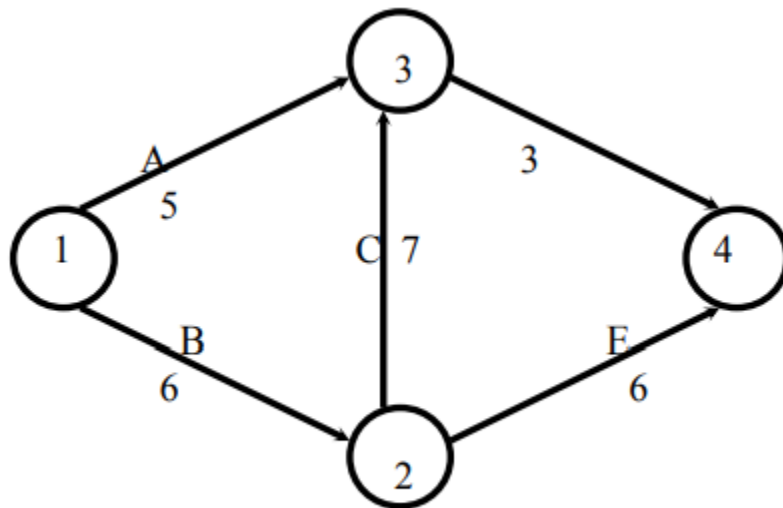


Fig. 12

We have five activities A, B, C, D and E with the time of completion of the activities 5, 6, 7, 3 and 6 days respectively. We represent the activities in a network shown in the same figure.

In this network, there are three ways to get from the starting point at node 1 and travel through the network to end at node 4. These ways are called paths. Thus, a path is defined as "a set of nodes connected by lines which begin at the initial node of a network and end at the terminal node". In figure12, there are three paths namely, 1-3-4, 1-2-3-4 and 1-2-4 where the numbers represent the nodes. The length of a path in a network is the total time it takes to travel along the path. This time is calculated by adding the individual times between the connected nodes on the path. A path is called a critical path if it is the longest the path in a project network. We have the times of the three distinct paths has shown below.

Path	Time (days)
1-3-4	$5 + 3 = 8$ days
1-2-3-4	$6 + 7 + 3 = 16$
1-2-4	$6 + 6 = 12$

The path connecting the nodes 1,2,3 and 4 constitutes the longest path and hence 1-2-3-4 is the critical path. The minimum time to complete the project is the time taken for the longest path

namely 16 days. The activities B (1- 2), C (2-3) and D(3-4) constitute the critical activities. These jobs are critical in determining the project's duration. The critical path for the above example can be shown in the diagram.

The same can be calculated using Activity On Node diagram. The AON diagram is as shown in figure 13.

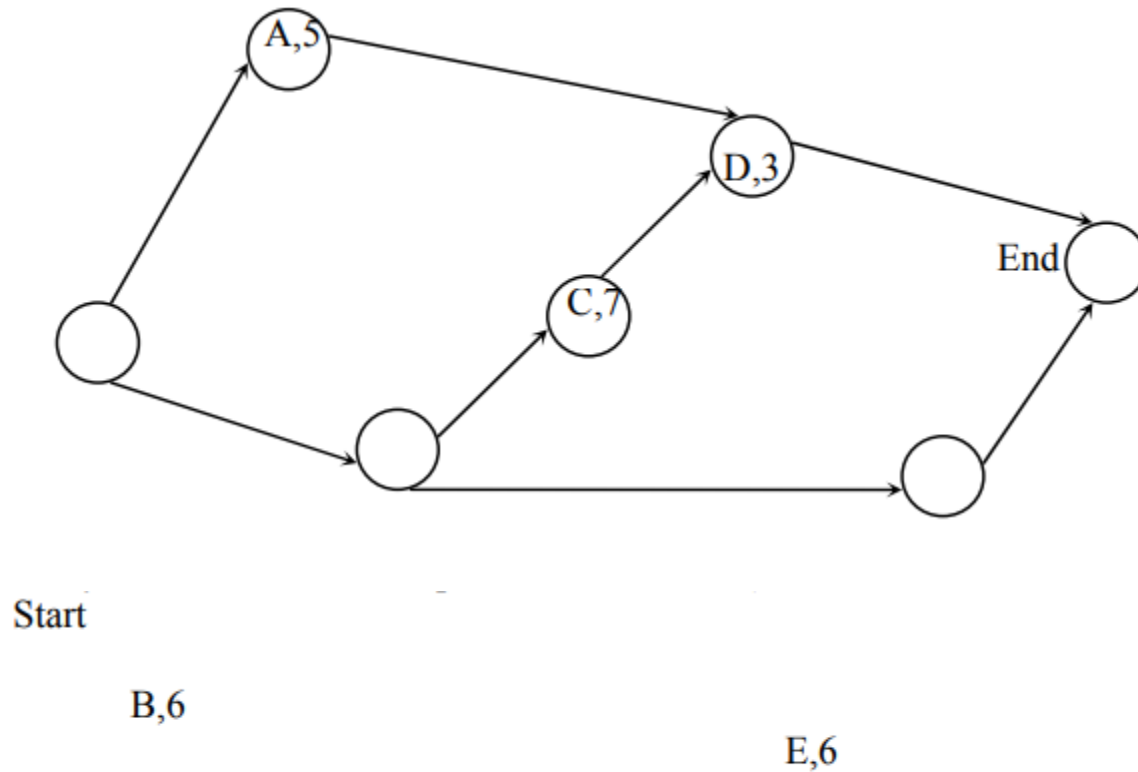


Fig. 13

We see that there are three paths or routes from start to end.

The following are the paths.

- (1) Start - A - D - End
- (2) Start - B - C - D - End
- (3) Start - B - E - End.

The times taken to complete the activities on the three paths are

$$5 + 3 = 8 \text{ days}$$

$$6 + 7 + 3 = 16 \text{ days}$$

$$6 + 6 = 12 \text{ days}$$

The longest path is B-C-D and the duration is 16 days. Hence the critical path is B-C-D and the project completion time is 16 days.

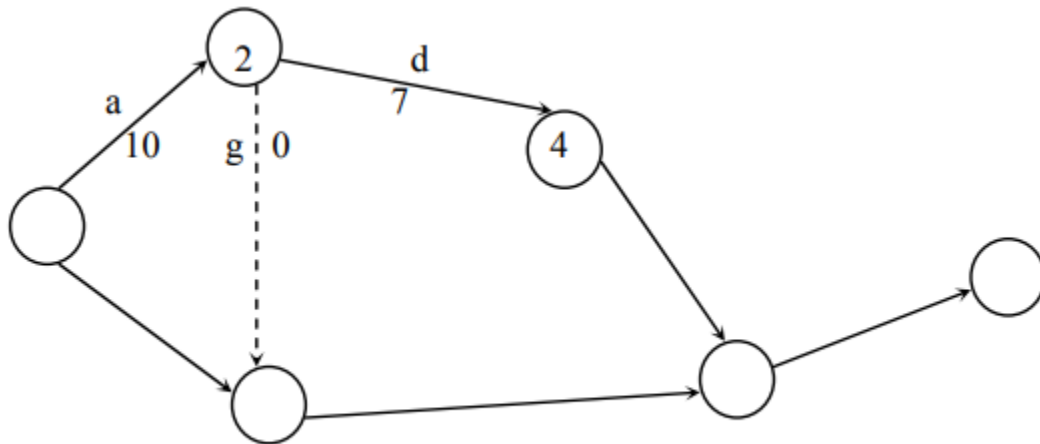
Thus, whether we use the arrow diagram or the activity on node diagram, we get the same result. This procedure can be applied without much effort for small projects involving only a few activities but the same procedure may be cumbersome if the project involves a large number of activities. However, there is an effective method of finding the critical path as explained in the next section.

ALGORITHM FOR CRITICAL PATH

Consider the following project.

Job's name	Immediate predecessor	Time to complete the job
a	-	10 days
b	-	3 days
c	a,b	4 days
d	a	7 days
e	d	4 days
f	c,e	12 days

We draw the project graph by an arrow diagram as shown in Fig 14. The activity 'g' is a dummy activity consuming no time.



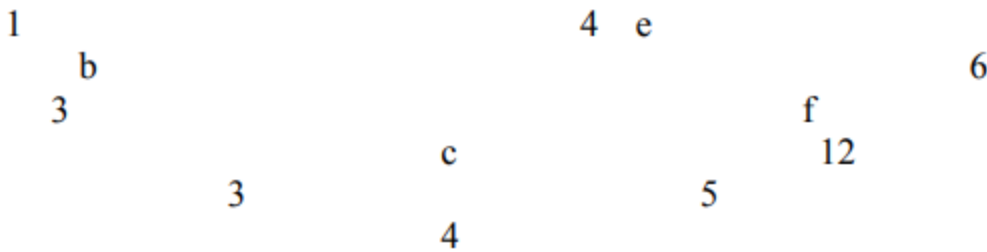


Fig. 14

Early start and early finish programme

We define the 'early start' of a job in a project as the earliest possible time when the job can begin. This is the first number in the bracket. The early finish time of a job is its early start time plus the time required to perform the job. We put this as the second number in the same bracket. Thus if in fig. 15 the early start of the job a will be 0 as this is the first job. The early finish $0 + 10 = 10$. For convenience we denote the start time of a project as 0. So we have early start and early finish times as (0, 10) for job a. For job b, the early start time will also start and the early finish times are (0, 3). The jobs d and g can start, at the earliest, as soon as the job a is completed, because 'a' is the predecessor for d and g. Hence the early start time for job d will be 10 and early finish time for job d will be $10 + 7 = 17$. For the job g the early start and early finish will (10, 10) since g happened to be a dummy job. Next we take the job c which has two predecessors namely b and g. The job c can be started, at the earliest on 10. So the job c can start, at the earliest only on 10 and the early finish of the job c will be $10 + 4 = 14$. Similarly the early start of job e is 17 and finish is $17 + 4 = 21$. The job f has two predecessors namely c and e, c is finished at the earliest on 14 and e on 21. So, the job f can start at the earliest only on 21 and finished at the earliest on 33.

All the above data are recorded as shown in figure 15.

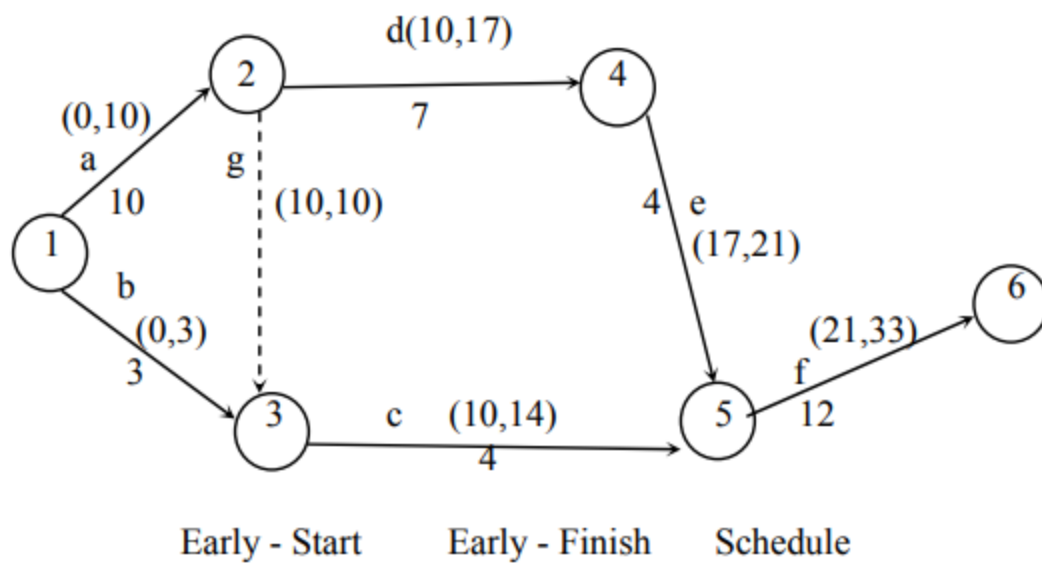


Fig. 15

Thus the project can be completed in 33 days. If the earliest completion time of the project is 33 days after it has begun, the longest path through the network must be 33 days in length. When we compare the alternate paths available for completion of the project (a-d-e-f = 33; a-g-c-f = 26; b-c-f = 19) from the beginning to the end we will find that the critical path is 33 days, with the jobs a, d, e, f on the critical path.

Late start and Late finish programme

The activities not on the critical path can be delayed without delaying the completion date of the project. A normal question may arise at this juncture as to how much delay can be allowed for non-critical jobs. How late can a particular activity be started and still maintain the length of the project duration? For answering these questions we find the late start and late finish times for each activity in the project graph.

We define the late start of an activity as the latest time that it can begin without pushing the finish date of the project further into the future. Similarly late finish of an activity is the late start time plus the activity duration.

We have seen that the early start and early finish time of activities are calculated in the forward direction from left to right. To calculate the late start and late finish times, we begin at the end of the network and work backwards. In the example cited above we begin with the node 6. The job leading to the node is job f. It must be completed by day 33 so as not to delay the project.

Therefore, day 33 is the latest finish. In general duration of the critical path will be taken as the late finish time of the project.

The time taken for the activity f is 12 days, so that it must begin at day 21 and end on 33. Thus the late start of the activity f is day 21.

We represent the late start and late finish times of the activity f within brackets (21, 33) in figure 16. The job f has two predecessors namely c and e. The late-finish times for both jobs can be taken as day 21. If this is so, c's late start will be $21 - 4$ or day 17 and e's late start will be $21 - 4$ or day 17. We can record the late start and late finish times for jobs c and e in the figure within brackets.

Now, the job d is the only predecessor to e and d can finish as late as day 17 which is the late start of e, so d can start as late as on day 10. Coming to the activity g, which is the predecessor to c, the late finish time of job g is the same as the late start time of c. So the late start and late finish time of g will be (17, 17). Since b precedes c, b also can finish as late as on day 17 and start as late as on day 14.

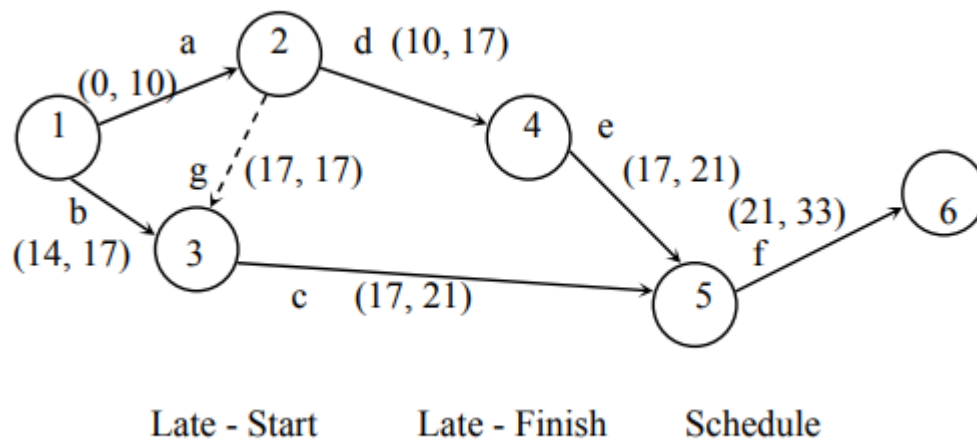


Fig.16

Next we take job a, which has two successors namely d and g, which have days 10 and 17 as the late start times respectively. So the job a must finish as late as on day 10 in order that a can be made to start on day 10. The finish time of the activity a will be least of the late start time of the successor activities. Hence the late finish of the activity a must be day 10 and it starts on day 0. This is also shown in figure 16.

The above information of early start-early finish programme and late start-late finish programmes is very useful as we shall see next.

Slack or Float

In the figure 15 and figure 16 which represent the early start-early finish and late start-late finish programmes, we observe that the late start and early start times are identical for some jobs. (Similarly late and early finish times). For example for the job a, the early and late start time is 0, for the job d, early and late start time is 10, for e they are 17 and for f also they are 21, whereas for job b the early start is day 0 and late start is day 14. This indicates that the job b can start as early as day 0 or as late as on day 14, without delaying the project completion date of 33 days. This difference between late start of the job and the early start of the same job is called as 'slack' or 'float'. It is also termed as 'total slack' or 'total float'. This denotes the maximum delay that can be allowed for this job. Similarly for job g, the early start time is day 10 and late start time is 17. There is a difference of 7 days which is the slack for job 'g'. Similarly we have a slack of 7 days for the job 'c' also.

The jobs having no slack thus become critical and the jobs with slacks are non-critical. So if we connect the jobs having no slacks, we get the critical path. This is the method of finding the critical path. In the example the jobs a, d, e, f have no slack and the path connecting a, d, e and f or 1-2-4-5-6 becomes the critical path.

We can summaries the slack for all jobs given in the tabular below.

Jobs	Early start	Early finish	Late start	Late finish	Slack
a	0	10	0	10	0
b	0	3	14	17	14
c	10	14	17	21	0
d	10	17	10	17	0
e	17	21	17	21	0
f	21	33	21	33	0
g	10	10	17	17	7

This critical path is shown in figure 17

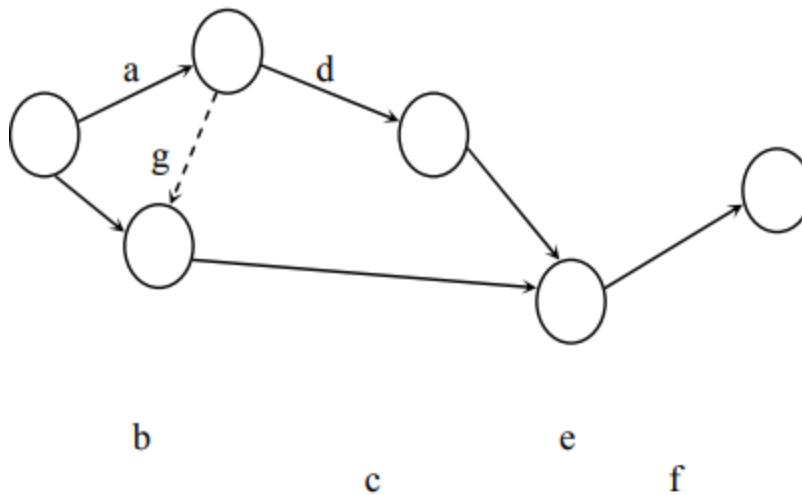


Fig. 17

Free Slack:

Free Slack is defined as the amount of time an activity can be delayed without affecting the early start time of any other job. In other words, the free slack of any activity is the difference between its early finish time and the earliest of the early start times of all its immediate successors. In the example above, if we take the activity b, its early finish time is day 3 and its immediate successor c starts at the earliest on day 10. 'c' can start as late as on day 17, but there is not compulsion on c to start exactly on day 10. If he chooses to start on day 10 (earliest start time) then b will have only 7 days as free slack (difference between early start of c and early finish of b). Similarly for the job c also we have free slack as the difference between the early finish of c and early start of its immediate successor 'f' i.e. $21 - 14 = 7$ days. Free slack can never exceed total slack. The total slack and free slack for all activities are given in the following table.

Activity	Total Slack	Free Slack
a	0	0
b	14	7
g	7	0
c	0	7
d	0	0
e	0	0
f	7	0

Independent slack: It is that portion of the total float within which an activity can be delayed for start without affecting slacks of the preceding activities. It is computed by subtracting the tail event slack from the free float. If the result is negative it is taken as zero.

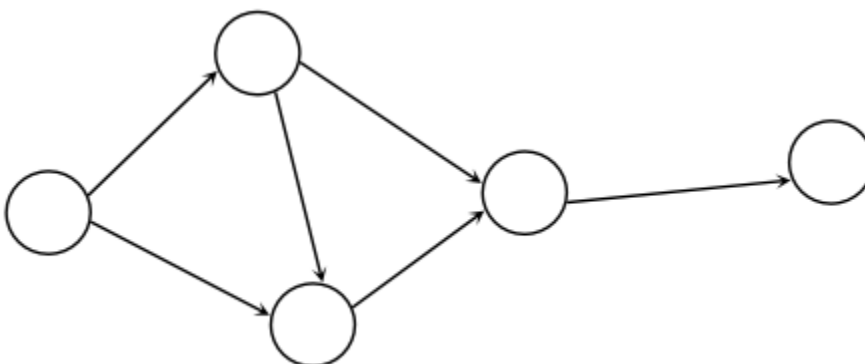
Example The following table gives the activities of a construction project and duration.

Activity	1-2	1-3	2-3	2-4	3-4	4-5
Duration (days)	20	25	10	12	6	10

- Draw the network for the project.
- Find the critical path.
- Find the total, free and independent floats each activity.

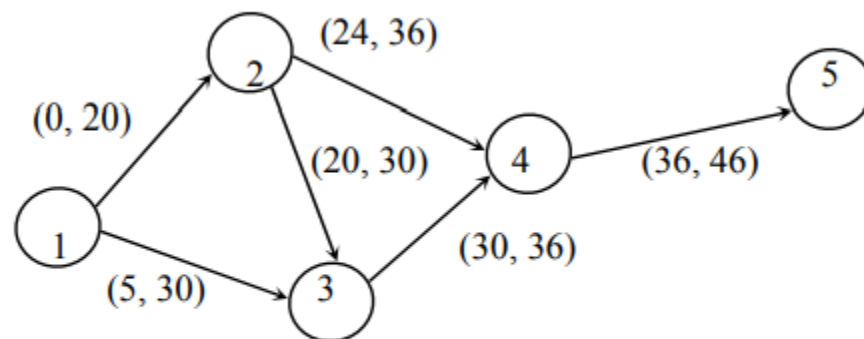
Solution:

The first step is to draw the network and fix early start and early finish schedule and then late start-late finish schedule as in figure 18 and figure 19.



2 (20, 32)
 (0, 20) 20 12
 10 (20, 30) 4 10 5
 1 25 (36, 46)
 (0, 25) 3 6 (30, 36)

Early-Start Early-Finish Schedule
Fig. 18



Late-Start Late-Finish Schedule

Fig. 19

Activity	Total Slack	Free Slack	Independent Slack
1-2	0	0	0
1-3	5	5	5
2-3	0	0	0
2-4	4	4	4
3-4	0	0	0
4-5	0	0	0

To find the critical path, connect activities with 0 total slack and we get 1-2-3-4-5 as the critical path.

Check with alternate paths.

$$\begin{array}{ll} 1-2-4-5 & = 42 \\ 1-2-3-4-5 & = 46^* \quad (\text{critical path}) \\ 1-3-4-5 & = 41 \end{array}$$

5PERT MODEL

PERT was developed for the purpose of solving problems in aerospace industries, particularly in research and development programmes. These programmes are subject to frequent changes and as such the time taken to complete various activities are not certain, and they are changing and non-standard. This element of uncertainty is being specifically taken into account by PERT. It assumes that the activities and their network configuration have been well defined, but it allows for uncertainties in activity times. Thus the activity time becomes a random variable. If we ask an engineer, or a foreman or a worker to give a time estimate to complete a particular task, he will at once give the most probable time required to perform the activity. This time is the most likely time estimate denoted by t_m . It is defined as the best possible time estimate that a given activity would take under normal conditions which often exist.

But he is also asked to give two other time estimates. One of these is a pessimistic time estimate. This is the best guess of the maximum time that would be required to perform an activity under the most adverse circumstances like

- (1) supply of materials not in time
- (2) non-cooperation from the workers
- (3) the transportation arrangements not being effective etc.

Thus the pessimistic time estimate is the longest time the activity would require and is denoted by t_p . On the other hand if everything goes on exceptionally well or under the best possible conditions, the time taken to complete an activity may be less than the most likely time estimate. This time estimate is the smallest time estimate known as the optimistic time estimate and denoted by t_o .

Thus, given the three time estimates for an activity, we have to find the expected duration of an activity or expected time of an activity as a weighted average of the three time estimates. PERT makes the assumption that the optimistic and pessimistic activity (t_o and t_p) are occur. It also assumes that the most probable activity time t_m , is four times more likely to occur than either of the other two. This is based on the properties of Beta distribution. Beta distribution was chosen

as a reasonable approximation of the distribution of activity times. The Beta distribution is unimodal, has finite non-negative end points and is not necessarily symmetrical-all of which seen desirable properties for the distribution of activity times. The choice of Beta distribution was not based on empirical data. Since most activities in a development project occur just once, frequency distribution of such activity times cannot be developed from past data.

So, if we follow Beta distribution with weights of 1, 4, 1 for optimistic, mostly likely and pessimistic time estimates respectively a formula for the expected time denoted by t_e can be written as

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

For example if we have 2, 5 and 14 hours as the optimistic (t_o), most likely (t_m) and pessimistic (t_p) time estimate then the expected time for the activity would be

$$\begin{aligned} t_e &= \frac{2+4(5)+14}{6} \\ &= \frac{36}{6} = 6 \text{ hrs} \end{aligned}$$

If the time required by an activity is highly variable (i.e) if the range of our time estimates is very large, then we are less confident of the average value. We calculate them if the range were narrower. Therefore it is necessary to have a means to measure the variability of the duration of an activity. One measure of variability of possible activity times is given by the standard deviation of their probability distribution.

PERT simplifies the calculation of standard deviation denoted by S_t as estimated by the formula,

$$S_t = \frac{t_p - t_o}{6}$$

S_t is one sixth of the difference between the two extreme time estimates, namely pessimistic and optimistic time estimates. The variance V_t of expected time is calculated as the square of the deviation.

$$(i.e.) \quad V_t = \left(\frac{t_p - t_o}{6} \right)^2$$

In the example above

$$S_t = \frac{14-2}{6} = 2 \text{ hrs}$$

$$V_t = \frac{(14-2)^2}{6^2} = 4 \text{ hr}^2$$

Activat

Expected length of a critical path:

The expected length of a sequence of independent activities is simply the sum of their separate expected lengths. This gives us the expected length of the entire project. We have to calculate the expected length t_e of every activity with the weights attached to the three time estimates and find the critical path in the manner described previously. The expected length of the entire project denoted by T_e is the length of the critical path (i.e.) the sum of the t_e 's of all activities along the critical path.

In the same way, the variance of a sum of independent activity times is equal to the sum of their individual variances. Since T_e is the sum of t_e 's along the critical path, then variance of T_e equals the sum of all the variances of the critical activities. The standard deviation of the expected project duration is the square root of the variance T_e as calculated above.

At this juncture, consider the following example to illustrate the application of these formulae.

Example

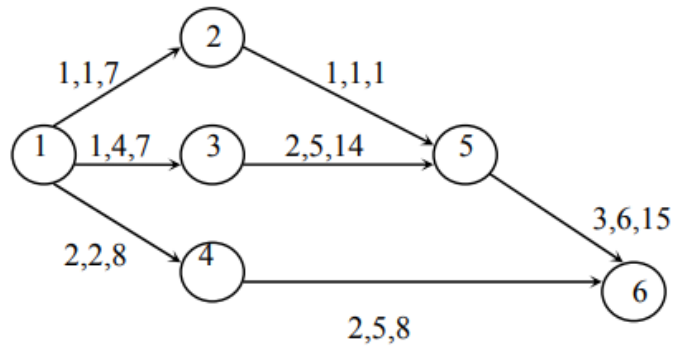


Fig. 20

Activity	Expected time (te)	Std. deviation	Variance
1-2	$(1+4+7)/6 = 2$	$(7-1)/6 = 1$	1
1-3	$(1+16+7)/6 = 4$	$(7-1)/6 = 1$	1
1-4	$(2+8+8)/6 = 3$	$(8-2)/6 = 1$	1
2-5	$(1+4+1)/6 = 1$	$(1-1)/6 = 0$	0
3-5	$(2+20+14)/6 = 6$	$(14-2)/6 = 2$	4
4-6	$(2+20+8)/6 = 5$	$(8-2)/6 = 1$	1
5-6	$(3+24+15)/6 = 7$	$(15-3)/6 = 2$	4

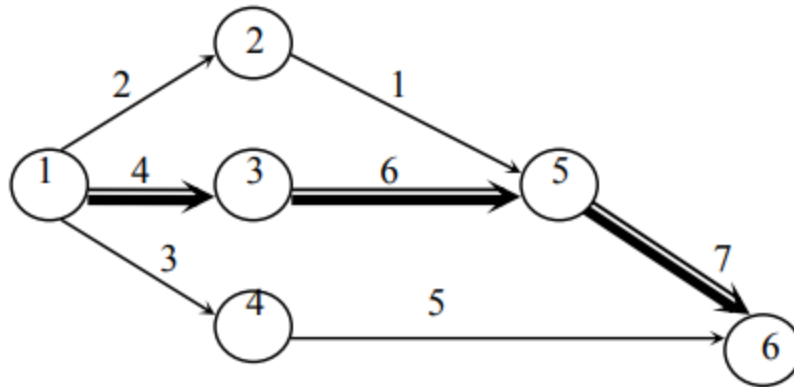
For each activity, the optimistic mostly likely and pessimistic time estimates are labeled in the same order. Using PERT formulae for t_e and St tabulate the results as above.

To calculate the critical path, list all the three paths with their expected time of completion from figure 21.

$$1-2-5-6 = 10$$

$$1-3-5-6 = 17$$

$$1-4-6 = 8$$



Expected time and Critical path.

Fig. 21

The longest path is the critical path and it consists of activities 1-3, 3-5 and 5-6 with a length $T_e = 4 + 6 + 7 = 17$. The variance of the critical path is then $V_T = \text{Variance of 1-3} + \text{variance of 3-5} + \text{variance of 5-6}$.

$$V_T = 1 + 4 + 4 = 9$$

Standard deviation of duration of the critical path

$$\sigma = \sqrt{V_T} = \sqrt{9} = 3$$

So far what we have done with PERT model is to recognize uncertainty by using three time estimates and these are reduced to a single time estimate for finding the critical path. It can be used to find early start-early finish programme, late start-late finish programme and slack. The variability of the time estimates for each activity is also reduced to a standard deviation and variance and this is used to find the standard deviation of expected completion time for the project.

Probability of completing a project with a given date

One may wonder how the calculations made to find T_e , ST and V_T are useful to a project manager. These parameters serve as a very useful tool for a project manager to estimate the probability of completing a project with a given date.

We have seen that the time required for an activity is uncertain and hence it is a random variable. Its average or expected value (t_e) can be estimated on the basis of an assumption regarding the type of probability distribution, and three points on this distribution namely optimistic, most likely and pessimistic time estimates.

We estimate the average, or expected, projected length, T_e by adding the expected activity durations along the critical path. As the t_e 's are all random variables, then so is T_e . It is assumed that T_e is to follow approximately a normal distribution. This gives the probability of completing a project with a given date.

Referring to the example illustrated above, the expected project length is 17 days and a standard deviation of 3 days. These are the two parameters useful to calculate the probability if the due date is to be met. For example, if the due date is T and this is deviated from the mean by $T - T_e$ and the same can be expressed as the ratio of standard deviation as $(T - T_e)/S.D.$ This is defined as the standard normal variates denoted by Z .

It is known that in a normal distribution the area under the normal curve gives the probability. For $Z = 0$, $T = T_e$ and hence if the due date is exactly the expected project length, the probability of completing the project is 50% as represented by the area to the left of the central line.

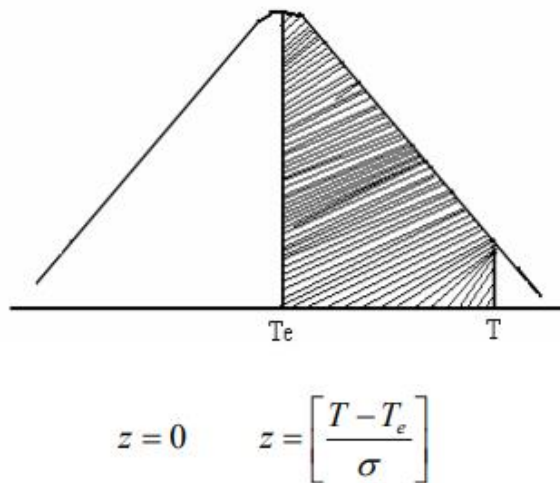


Fig. 22

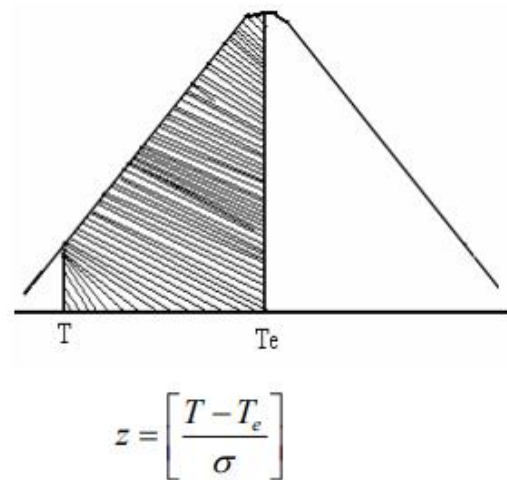


Fig 23

If in this example the due date is 20 days, $Z = (20 - 17)/3 = 1$. For the value of $Z = 1$ the area between $Z = 0$ and $Z = 1$ is estimated (or as found from the statistical tables) as 0.3413. This is

indicated in figure 222 by the shaded portion. Hence the probability of completing the project in 20 days will be $0.50 + 0.3413 = 0.8413$ or 84.13%. Similarly if the due date is 14 days, the corresponding value of $Z = -1$. For this value of $Z = -1$ also, area is 0.3413 but to the left of the mean as indicated in figure 23.

The probability of completing the project in 14 days will be $= 0.5000 - 0.3413 = 0.1587$ or 15.87%.

Example The following table lists the jobs of a network with their time estimates.

Job		Duration (days)		
i-j		Optimistic	Most likely	Pessimistic
1-2		3	6	15
1-6		2	5	14
2-3		6	12	30

2-4	2	5	8
3-5	5	11	17
4-5	3	6	15
6-7	3	9	27
5-8	1	4	7
7-8	4	19	28

- Draw the project network.
- Calculate the length and variance of the critical path.
- What is the approximate probability that the jobs on the critical path will be completed by the due date of 42 days?
- What due date has about 90% chance of being met?

Solution:

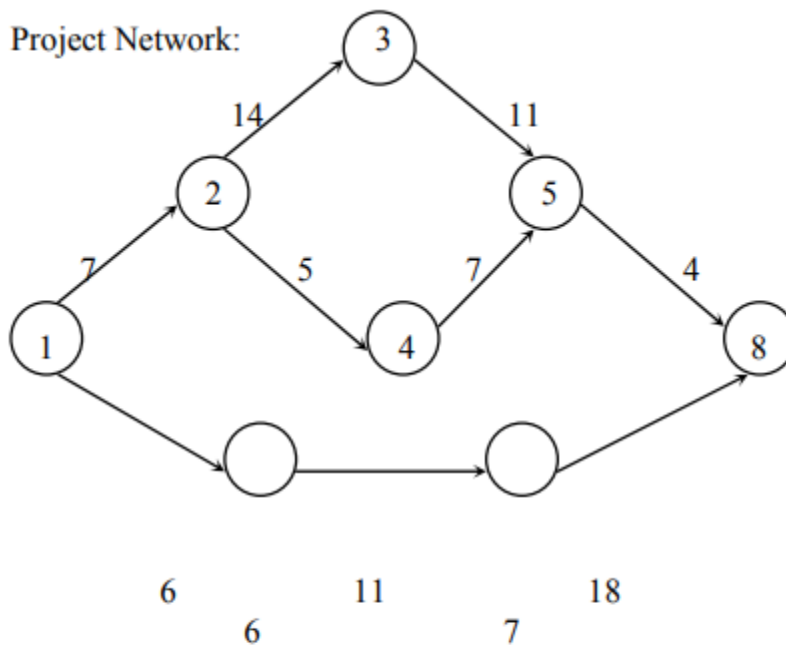
Before proceeding to draw the project network, let us calculate the expected time of activity t_e , standard deviation and variance of the expected time of activity using

$$t_e = \frac{(t_o + 4t_m + t_p)}{6}$$

$$S.D = (t_p - t_o) / 6; \quad \text{Variance} = (S.D)^2$$

Activity	t_e (Days)	S.D (Days)	Variance
1 - 2	7	2	4
1 - 6	6	2	4
2 - 3	14	4	16
2 - 4	5	1	1
3 - 5	11	2	4
4 - 5	7	2	4
6 - 7	11	4	16
5 - 8	4	1	1
7 - 8	18	4	16

(a) Project Network:



Expected time and Critical path
Fig. 24

(b) There are three paths:

$$1-2-3-5-8 = 36 \text{ days}$$

1-2-4-5-8 = 23 days

1-6-7-8 = 35 days

1-2-3-5-8 is the longest path and hence the critical path.

Expected length of the critical path is 36 days. The variance for 1-2, 2-3, 3-5 and 5-8 are 4, 16, 4 and 1 respectively and variance of the projection duration is 25 and hence

Standard deviation of the project duration = $\sqrt{25} = 5$ days.

(c) Due date = 42 days (T)

Expected duration = 36 days (T_e) and S.D = 5 days (o)

$$Z = (T - T_e) / S.D. = (42 - 36) / 5 = 1.2$$

The area under the normal curve for $Z = 1.2$ is 0.3849.

Therefore, the probability of completing the project in 42 days

= $0.5000 + 0.3849$

= 0.8849

= 88.49%

Exercise

(1) A project has the following characteristics:

Activity	Optimistic time	Pessimistic time	Most likely time
1-2	1	5	1.5
2-3	1	3	2
2-4	1	5	3
3-5	3	5	4
4-5	2	4	3
4-6	3	7	5
5-7	4	6	5
6-7	6	8	7
7-8	2	6	4

7-9	5	8	6
8-10	1	3	2
9-10	3	7	5

Construct a network. Find the critical path and variance for each event. Find the project duration at 95% probability.

RESOURCE SCHEDULING IN NETWORK

In scheduling projects using PERT or CPM, we have assumed that the only constraint for an activity is to fix the starting date and the finishing date. An activity may be started as soon as all its predecessors have been completed. Thus we have an early start schedule for all the activities. In such scheduling procedures it is assumed that the resources required to perform activities are available to any unlimited extent or atleast sufficient resources are available for the activity to be started sometime between earliest and latest start dates. But we come across situations where the project managers have to face the problem of limited or fixed resources like manpower availabilities, machine availabilities and capital mobilization. Activities, which occur on parallel paths through network, may compete for the same resource.

Recently considerable work is in progress in developing heuristic programmes for solving large combinatorial problems, A heuristic can be defined as a guide or method of reducing search in a problem-solving situations. It gives a rule of thumb. Businessmen employ heuristics quite often for their problems. As an example we consider the classic management rule, "**Handle only exceptional problems and let the subordinates decide on routine matters**" as a heuristic. Sometime a simple rule of thumb may not be enough, but other rules are also to particular problem is referred to as 'heuristic programme' requiring a computer for complex problems. **A number of different heuristic programmes for scheduling projects with limited resources have been developed recently.** These programmes are classified in any one of the two following forms.

1. Resource leveling programmes: These programmes seek an attempt to reduce peak resource requirements and smooth out period to period assignments with a restriction on project duration. In other words, in these programmes the resource requirements (manpower or money power) are kept at a constant level by altering the start time of the different activities. But at the same time, the project duration and the sequence are not violated.
2. Resource allocation programmes: These programs make an attempt to allocate the available resources (manpower, machine hours, money) to project activities and to find the shortest project schedule consistent with fixed resource limits. The following articles deal with these types of heuristic programmes.

Resource leveling program: Resource leveling programmes are most appropriate to situations where we maintain a relatively stable employment levels and utilize the resources at a more constant rate. If the levels change either way the expenses are involved in hiring, training, firing, unemployment insurance and so on. The activity slack is a measure of flexibility in the assignment of activity start times. We may use activity slack as a means to smoothing peak resource requirements. The procedure involves preparing an early-start early-finish schedule for the project on time scale and then plotting a resource-loading chart for the schedule. The activities are shifted suitably to reduce the peak resource requirements without violating the sequence and preserving the finish date.

Resource Allocation: In many cases the scheduling problem does not involve only smoothing of requirements but also allocation of resources. Almost there are some constraints, which may be in the form of limited men, machines or finance. The various activities have to work at their optimum taking into considerations these restrictions.

The resource-leveling model could be modified to give us the desired results. The constraint of the fixed due date would have to be removed and concentrate on pushing down the trigger levels until peak requirements are all under resource limits. If the limits are confining, the jobs will be pushed to the right, thereby delaying the project due date on the schedule graph. The constraints cannot be based both on project due date and resource limits.

Resources are allocated on a period-by-period basis to accommodate some available job, after their predecessors have been completed. The essential elements are determining which jobs are to be scheduled and which are to be postponed. The slack is usually the basis on which priority is accorded to various job. Hence, the jobs, which are most critical, will be scheduled first.

In any problem on resource allocation, the following three rules may be applied.

- (i) The resources are to be allocated in the order of time i.e. start on the first day and schedule all possible jobs then the second day etc.
- (ii) When several jobs compete for the same resources preference should be given to the job with least slack.
- (iii) The non-critical jobs should be rescheduled so that resources would be free for scheduling the critical jobs.

Exercises

1. Following are the manpower requirements for each activity in a project.

Activity	Normal Time (days)	Manpower required per day
1 - 2	10	2
1 - 3	11	3
2 - 4	13	4
2 - 6	14	3
3 - 4	10	1
4 - 5	7	3
4 - 6	17	5
5 - 7	13	3
6 - 7	9	8
7 - 8	1	11

(i) Draw the network and find 'total float' and 'free float' for each activity.

(ii) The contractor stipulates that during the first 26 days only 4 to 5 men and during the remaining days 8 to 11 men only can made be available. Rearrange the activities for leveling the manpower resources satisfying the above condition.

2. For a project consisting of several activities, the duration and required resources for carrying out each of the activities and their availabilities are given below. Find the project completion time under the given resource constraints.

Activity	Resources Required		Duration (days)
	Equipment	Operators	
1 - 2	X	30	4
1 - 3	Y	20	3
1 - 4	Z	20	6
2 - 4	X	30	4
2 - 5	Z	20	8
3 - 4	Y	20	4
3 - 5	Y	20	4
4 - 5	X	30	6

Resource availability: Number of operators = 50

Equipments $X = 1, Y = 1, Z = 1$

Segment III: Inventory Control

Lectures 11 -16

INVENTORY AND CONTROL

Inventory is the physical stock of items held in any business for the purpose of future production or sales. In a production shop the inventory may be in the form of raw materials. When the items are in production process, we have the inventory as in-process inventory and at the end of the production cycle inventory is in the form of finished goods. We shall be dealing only with the finished goods inventory. The problem of determining inventory policies is not a new concept beginning. It is only in the last two decades that it has been tackled with quantitative techniques and mathematical models, a method amenable to optimization. Inventory planning is the determination of the type and quantity of inventory items that would be required at future points for maintaining production schedules. Inventory planning is generally based on information from the past and also on factors that would arise in future. Once this sort of planning is over, the control process starts, which means that actual and planned inventory positions are compared and necessary action taken so that the business process can function efficiently.

In inventory control, we are primarily concerned with the inventory cost control. The aim is focused to bring down the total inventory cost per annum as much as possible. Two important questions are (1) *how much to stock or how much to buy* and (2) *how often to buy or when to buy*. An answer to the above questions is usually given by certain mathematical models, popularly known as ‘economic order quantity models’ or ‘economic lot/batch size models (E.O.Q.).’

INVENTORY COSTS

There are four major elements of inventory costs that should be taken for analysis, such as

- (1) Item cost, Rs. C_1/item .
- (2) Ordering cost, Rs. C_2/order .
- (3) Holding cost Rs. $C_3/\text{item/unit time}$.
- (4) Shortage cost Rs. $C_4/\text{item/Unit time}$.

Item Cost (C_1)

This is the cost of the item whether it is manufactured or purchased. If it is manufactured, it includes such items as direct material and labour, indirect materials and labour and overhead expenses. When the item is purchased, the item cost is the purchase price of 1 unit. Let it be denoted by Rs. C_1 per item.

Purchasing or Setup or Acquisition or Ordering Cost (C_2)

Administrative and clerical costs are involved in processing a purchase order, expediting, follow up etc., It includes transportation costs also. When a unit is manufactured, the unit set up cost includes the cost of labour and materials used in the set up and set up testing and training costs. This is denoted by Rs. C_2 per set up or per order.

Inventory holding cost (C_3)

If the item is held in stock, the cost involved is the item carrying or holding cost. Some of the costs included in the unit holding cost are

- (1) Taxes on inventories,
- (2) Insurance costs for inflammable and explosive items,
- (3) Obsolescence,
- (4) Deterioration of quality, theft, spillage and damage to times,
- (5) Cost of maintaining inventory records.

This cost is denoted by Rs. C_3 /item/unit time. The unit of time may be days, months, weeks or years.

Shortage Cost (C_4)

The shortage cost is due to the delay in satisfying demand (due to wrong planning); but the demand is eventually satisfied after a period of time. Shortage cost is not considered as the opportunity cost or cost of lost sales.

The unit shortage cost includes such items as,

- (1) Overtime requirements due to shortage,
- (2) Clerical and administrative expenses.
- (3) Cost of expediting.
- (4) Loss of goodwill of customers due to delay.
- (5) Special handling or packaging costs.
- (6) Lost production time.

This cost is denoted by Rs. C_4 per item per unit time of shortage.

INVENTORY MODELS (E.O.Q. MODELS)

The inventory control model can be broadly classified into two categories:

- (1) Deterministic inventory problems.
- (2) Probabilistic or stochastic inventory problems.

In the deterministic type of inventory control, the parameters like demand, ordering quantity cost, etc are already known or have been ascertained and there is no uncertainty. In the stochastic inventory control, the uncertain aspects are taken into account.

First let us consider the inventory control of the deterministic type. There are four EOQ models which are discussed below. The first one is the well-known Wilson's inventory model.

Model 1: Purchasing model with no shortages: (Wilson's model)

The following assumptions are made in deriving the formula for economic order quantity.

- (1) Demand (D) is at a constant rate.
- (2) Replacement of items is instantaneous (lead time is zero).
- (3) The cost coefficients C_1 , C_2 , and C_3 are constant.
- (4) There is no shortage cost or $C_4 = 0$.

This mode represented graphically in fig. 1. This is also known as a saw tooth model (because of its shape).

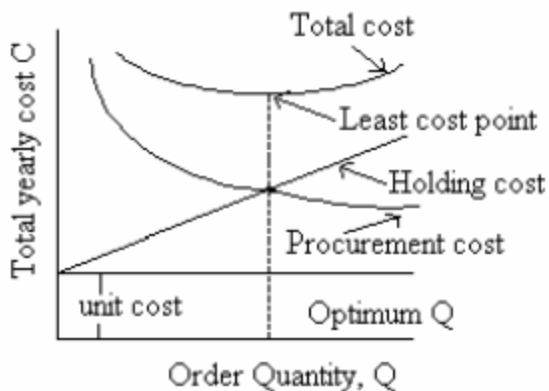
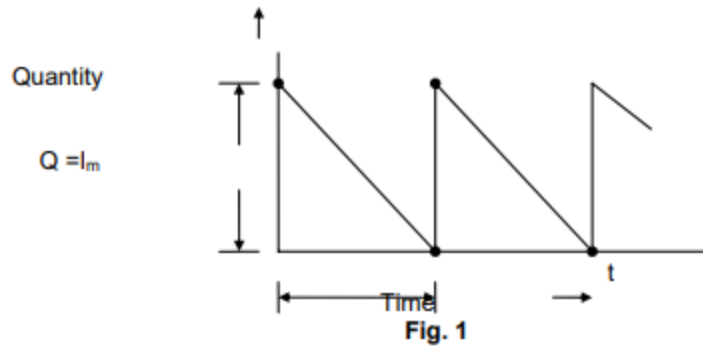


Fig. 2

In this model, at time $t = 0$, we order a quantity Q which is stored as maximum inventory, I_m . The time ' t ' denotes the time of one period or it is the time between orders or it is the cycle time. During this time, the items are depleting and reaching a zero value at the end of time t . At time t another order of the same quantity is to be placed to bring the stock upto Q again and the cycle is repeated. Hence this is a fixed order quantity model.

The total cost for this model for one cycle is made up of three cost components.

$$\text{Total cost/period} = (\text{Item cost} + \text{set up cost} + \text{holding cost/period})$$

$$\begin{aligned} \text{Item cost per period} &= (\text{Cost of item}) \times (\text{number of items ordered/period}) \\ &= C_1 Q \end{aligned} \quad (1)$$

$$\text{Purchase or set up cost per period} = C_2 \text{ (only one set up per period)}$$

$$\begin{aligned} \text{Item holding cost per period} &= (\text{Holding cost}) \times (\text{average inventory per period}) \times (\text{time per period}) \\ &= \times C_3 t Q/2 \times t \end{aligned} \quad (2)$$

Therefore the total cost per period $(C') = C_1Q + C_2 + C_3 \frac{Q}{2} \times t$ (3)

But the time for one period $t = Q/D$ (4)

Therefore the total cost per unit time, $C = C'/t$ (5)

$$C = C_1 + C_2 \frac{D}{Q} + C_3 \frac{Q}{2} \quad (6)$$

Substituting the value of t , we get $C = C_1 + C_2 \frac{D}{Q} + C_3 \frac{Q}{2}$ (7)

The cost components of the above equations can be represented as shown in fig 2 and an optimum order quantity for one period is found when

Purchase cost = Item holding cost.

$$\frac{C_2 D}{Q} = \frac{C_3 Q}{2}$$

$$Q_2 = \frac{2C_2 D}{C_3} \quad Q^* = \sqrt{\frac{2C_2 D}{C_3}} \quad (8)$$

This minimum inventory cost per unit time can also be found by differentiating C with respect to Q and equating it to zero. The derivative of the equation is,

$$\frac{dC}{dQ} = \frac{-C_2 D}{Q^2} + \frac{C_3}{2} = 0 \quad (9)$$

Solving for Q , we get

$$Q^* = \sqrt{\frac{2C_2 D}{C_3}} \quad (10)$$

This value of Q^* is the economic order quantity and any other order quantity will result in a higher cost.

The corresponding period t^* is found from

$$t^* = Q^*/D \quad (11)$$

The optimum number of orders per year is determined from

$$N^* = D/Q^*$$

where D is the demand per year.

Example 1: The demand rate for a particular item is 12000 units/year. The ordering cost is Rs. 100 per order and the holding cost is Rs. 0.80 per item per month. If no shortages are allowed and the replacement is instantaneous, determine:

- (1) The economic order quantity.
- (2) The time between orders.
- (3) The number of orders per year.
- (4) The optimum annual cost if the cost of item is Rs. 2 per item.
- (5)

Solution: Note that the holding cost is given per month and convert the same into cost per year.

$$C_1 = \text{Rs. 2/item}$$

$$C_2 = \text{Rs. 100/order}$$

$$C_3 = \text{Rs. 0.80/item/month}$$

$$= \text{Rs. 9.6/item/year}$$

$$D = 12000 \text{ items/year}$$

a) The economic order quantity

$$\begin{aligned} Q^* &= \sqrt{\frac{2C_2D}{C_3}} \\ &= \sqrt{\frac{2 \times 100 \times 12000}{9.6}} \\ &= 500 \text{ units} \end{aligned}$$

b) The time between orders

$$t^* = Q^*/D = 500 / 12000 \text{ yr.}$$

$$= 500/1000 \text{ month}$$

$$= 0.5 \text{ month}$$

c) The number of orders/year

$$N = D/Q^* = 12000 / 500 = 24$$

d) The optimum annual cost

$$= (\text{Number of orders}) \times (\text{Cost of one cycle})$$

$$= 24(500 \times 2) + 100 + (500 / 2 \times 0.5 \times 0.80)$$

$$= \text{Rs. } 28800$$

Model 2 Purchasing model with shortages

In this model, shortages are allowed and consequently a shortage cost is incurred. Let the shortages be denoted by 'S' for every cycle and shortage cost by C_4 per item per unit time. This model is illustrated in Fig .3

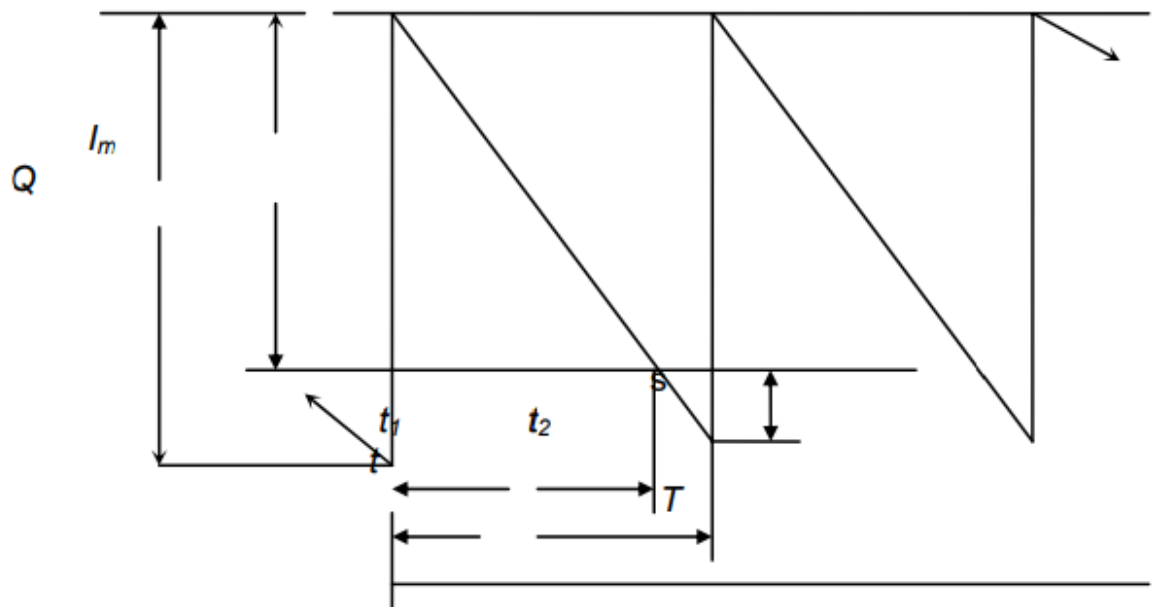


Fig. 3

Fig. 3 shows that the back ordering is possible (i.e.) once an order is received, any shortages can be made up as the items are received. Consequently shortage costs are due to being short of stock for a period of time.

The cost per period includes four cost components.

Total cost per period = Item cost + Order cost + Holding cost + Shortage cost

Item cost per period = (item cost) $\times$ (number of items/period)

$$= C_1 Q \quad (13)$$

Order cost per period = C_2

Let t_1 be the time period during which only the items are held in stock. Let the maximum inventory be denoted by I_m

and this is equal to $(Q - S)$ or $I_m = (Q - S)$

From similar triangle concept, the following equations can be obtained, referring to fig 3

$$t_1 / I_m = t / Q \quad (16)$$

or $t_1 = t I_m / Q = t(Q - S) / Q \quad (17)$

$$t_2 / S = t / Q \quad \text{or} \quad t_2 = t S / Q \quad (18)$$

Since time of one period $t = Q / D$

$$t_1 = \frac{Q - S}{Q} \frac{Q}{D} \quad (19)$$

$$t_2 = \frac{S}{Q} \frac{Q}{D} \quad (20)$$

Holding cost per period

= Average stock/period $\times t_1 \times$ holding cost/unit/unit time

$$= I_m / 2 \times t_1 \times C_3$$

$$= C_3 \times (Q - S) / 2 \times (Q - S) / Q \times Q / D$$

$$= \frac{C_3 (Q - S)^2}{2D} \quad (21)$$

Shortage cost per period

Average shortages/period $\times t_2 \times$ shortage cost

$$= S/2 \times S/Q \times Q/D \times C_4$$

$$= C_4 \times S^2/2D \quad (22)$$

Adding all the four cost components, we get the inventory cost per period.

$$C' = C_1Q + C_2 + \frac{C_3(Q-S)^2}{2D} + \frac{C_4S^2}{2D} \quad (23)$$

Therefore inventory cost per unit of time is obtained by dividing C' by t or Q/D

Therefore,

$$\begin{aligned} C &= \frac{C_1QD}{Q} + \frac{C_2D}{Q} + \frac{C_3(Q-S)^2D}{2D \times Q} + \frac{C_4S^2}{2D} \frac{D}{Q} \\ &= C_1D + \frac{C_2D}{Q} + \frac{C_3(Q-S)^2}{2Q} + \frac{C_4S^2}{2Q} \end{aligned} \quad (24)$$

This is an expression involving two variables Q and S . For optimum values of Q^* and S^* , the function has to be differentiated partially with respect to Q and S and equated to zero.

$$\frac{\partial C}{\partial Q} = 0$$

$$\begin{aligned} &= \frac{-C_2 D}{Q^2} + \frac{C_3}{2} - \frac{C_3 S^2}{2Q^2} - \frac{C_4 S^2}{2Q^2} \\ &= -\frac{C_2 D}{Q} + \frac{C_3}{2} - \frac{S^2}{2Q^2} (C_3 + C_4) \end{aligned} \quad (25)$$

$$\frac{\partial C}{\partial S} = 0$$

$$= -C_3 + \frac{C_3 S}{Q} + \frac{C_4 S}{Q} = -C_3 + \frac{S(C_3 + C_4)}{Q} \quad (26)$$

Solving the equation (26) for S, we get

$$S = \frac{C_3 Q}{C_3 + C_4} \quad (27)$$

Substituting the equation (27) into the equation (25), we get

$$0 = \frac{-C_2 D}{Q^2} + \frac{C_3}{2} - \frac{C_3 + C_4}{2Q^2} \times \frac{(C_3 + Q)^2}{(C_3 + C_4)} \quad (28)$$

$$= \frac{-C_2 D}{Q^2} + \frac{C_3}{2} - \frac{C_3^2}{2(C_3 + C_4)} \quad (29)$$

Solving equation (29) for Q, we get

$$Q^* = \sqrt{\frac{2C_2 D}{C_3}} \quad \sqrt{\frac{C_3 + C_4}{C_4}} \quad (30)$$

which is the economic or optimum order quantity.

$$S^* = \sqrt{\frac{2C_2D}{C_4}} \sqrt{\frac{C_3}{C_3+C_4}} \quad (31)$$

Example: The demand for an item is 18000 units/year. The cost of one purchase is Rs. 400. The holding cost is Rs. 1.2 per unit per year. The item cost is Rs. 1 per item. The shortage cost is Rs. 5 per unit per year. Determine:

- The optimum order quantity.
- The time between orders.
- The number of orders per year.
- The optimum shortages.
- The maximum inventory.
- The time of items being held.
- The optimum annual cost

Solution

$D = 18000$ units/year	or	1500 units/month
$C_1 = \text{Rs. } 1/\text{item}$		$C_3 = \text{Rs. } 1.2/\text{year/item}$
$C_2 = \text{Rs. } 400/\text{order}$		$C_4 = \text{Rs. } 5/\text{year/item}$

$$\begin{aligned} \text{a) } Q^* &= \sqrt{\frac{2C_2D}{C_3}} \sqrt{\frac{C_3 + C_4}{C_4}} = \sqrt{\frac{2 \times 400 \times 18000}{1.2}} \times \sqrt{\frac{6.2}{5}} \\ &= 3857 \text{ Units.} \end{aligned}$$

$$\text{b) } t^* = Q^*/D = 3857/1500 = 2.57 \text{ months}$$

$$\text{c) } \text{Number of orders per year} = 12/2.57 = 4.66$$

$$\text{d) } S^* = \sqrt{2C_2 D/C_4} \sqrt{C_3/(C_3+C_4)} = 747 \text{ items}$$

$$\text{e) } I_{\max} = Q^* - S^* = 3857 - 747 = 3110 \text{ items}$$

$$\text{f) } t_1 = I_{\max} / D = 3110 / 1500 = 2.07 \text{ months}$$

g) Annual cost = Item cost
 + Ordering cost
 + Holding cost
 + Shortage cost
 Item cost = Rs. 3857 per order
 Order cost = Rs. 400 per order

$$\text{Holding cost} = \frac{\text{Rs. } 1.2 \times 3110 \times 2.07}{2 \times 12}$$

$$= \text{Rs. } 322.14 \text{ per order}$$

$$\text{Shortage cost} = \frac{\text{Rs. } 5 \times 14 \times 0.5}{2 \times 12}$$

$$= \text{Rs. } 77.75 \text{ per order}$$

$$\begin{aligned} \text{Total cost per order} &= 3857 + 400 + 77.75 + 322.14 \\ &= \text{Rs. } 4656.89 \end{aligned}$$

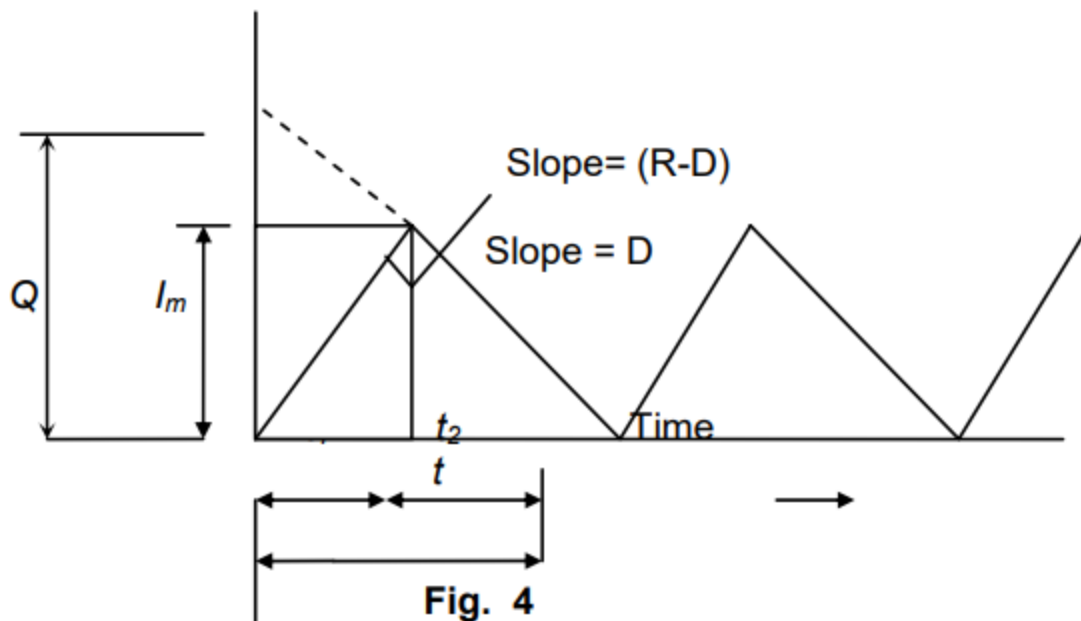
$$\begin{aligned} \text{Annual cost} &= \text{Number of orders/year} \times \text{cost} \\ &= 4.66 \times 4656.89 \\ &= \text{Rs. } 21701. \end{aligned}$$

Model 3: Manufacturing model with no shortages

In this model the following assumptions are made:

- (1) Demand is at a constant rate (D).
- (2) All cost coefficients (C_1 , C_2 , C_3) are constants.
- (3) There is no shortage cost, or $C_4 = 0$.
- (4) The replacement rate is finite and greater than the demand rate. This is also called replenishment rate or manufacturing rate, denoted by R.

Schematically, this model is illustrated in fig 4



The total cost of inventory per period is the sum of three components: item cost, order cost and items holding cost.

Let I_m be the maximum inventory, t_1 be the time of manufacture and t_2 be the time during which there is no supply.

In this model, all items required for a cycle are not stored at the beginning as in Wilson's Model. The items are manufactured at a higher rate than the demand so that the difference $(R-D)$ is the existing inventory till the items are exhausted.

$$\text{Item cost/period} = C_1 Q$$

(32)

$$\text{Order cost/period} = C_2$$

(33)

$$\text{Item holding cost/period} = C_3 \times I_m \times (t_1 + t_2) / 2 \quad (34)$$

$$= C_3 I_m \times t / 2 \quad (35)$$

$$I_m = t_1 (R - D) \quad (36)$$

$$\text{But} \quad t_1 = Q / R \quad (37)$$

$$\text{Therefore} \quad I_m = (Q / R)(R - D) \quad (38)$$

Substituting the value of I_m , we get the total cost of inventory per period.

$$C' = C_1 Q + C_2 + C_3 (Q / R)(R - D) \times t / 2 \quad (39)$$

Total cost of inventory per unit time

$$C = C' / t \quad (40)$$

$$= C_1 Q / t + C_2 / t + C_3 (Q / R)(R - D) \times t / 2 \quad (41)$$

$$\text{But } t = Q / D$$

Substituting the value of t we get

$$C = C_1 D + C_2 D / Q + C_3 (Q / R)(R - D) / 2 \quad (42)$$

Differentiating C with respect to Q and setting equal to zero for minimum C , we get,

$$\frac{dC}{dQ} = 0 - \frac{C_2 D}{Q^2} + \frac{C_3 (R - D)}{2R} = 0 \quad (43)$$

Solving equation (43), we get

$$Q^* = \sqrt{\frac{2C_2 D}{C_3}} \sqrt{\frac{R}{R - D}} \quad (44)$$

This gives the economic order quantity and is a balance between holding and set up costs.

Example: The demand for an item in a company is 18000 units/year and the company can produce at the rate of 3000 per month. The cost of one set up is Rs. 500 and the holding cost of 1 unit per month is 15 paisas. Determine:

- a) The optimum manufacturing quantity.
- b) The maximum inventory.
- c) The time between orders.
- d) The number of orders/year.
- e) The time of manufacture.
- f) The optimum annual cost if the cost of the item per unit is Rs. 2.

Assume no shortages.

Solution

$C_1 = \text{Rs. 2 per item.}$

$C_2 = \text{Rs. 500 per order.}$

$C_3 = \text{Rs. 0.15 per item per month}$

$D = 18000/\text{year} = 1500/\text{month}$

$R = 3000/\text{month}$

a) Optimum manufacture quantity

$$Q^* = \sqrt{\frac{2C_2D}{C_3}} \sqrt{\frac{R}{R-D}}$$
$$= \sqrt{\frac{2 \times 500 \times 1500 \times 3000}{0.15(3000-1500)}} = 4470 \text{ units}$$

b) The maximum inventory

$$I_m = Q(R-D)/R = 4470 \times 1500/3000 = 2235 \text{ units}$$

c) The time between orders

$$t = Q/D = 4470/1500 = 2.98 \text{ months}$$
$$= 3 \text{ months}$$

d) The number of orders/year

$$N = 12/3 = 4$$

e) The time of manufacture

$$t_1 = Q/R = 4470/3000 = 1.490 \text{ months}$$

f) The optimum annual cost

$$= \text{Item cost} + \text{Ordering cost} + \text{Holding cost}$$

$$\text{Item cost} = 18000 \times 2 = \text{Rs. } 36000$$

$$\text{Ordering cost} = 500 \times 18000 / 4470 = \text{Rs. } 2013.5$$

$$\begin{aligned} \text{holding cost} &= 0.15 \times 12 \times 2235 \times 18000 / 36000 \\ &= \text{Rs. } 2011.5 \end{aligned}$$

$$\begin{aligned} \text{Total annual cost} \\ &= 36000 + 2013.5 + 2011.5 \\ &= \text{Rs. } 40025.0 \end{aligned}$$

Model 4. Manufacturing model with shortages

The assumptions in this model are the same as in model 3 except that the shortages are also considered. This model is illustrated in the figure in power point presentation of the lecture, given on the next page.

There are four components of inventory costs in this model. They are

- (1) Item cost.
- (2) Set up or order cost.
- (3) Items holding cost.
- (4) Shortage cost.

The items t_1 , t_2 , t_3 and t_4 are as represented in figure on the next page. The total cost per period t is

$$C' = C_1 Q + C_2 + C_3 (t_1 + t_2) I_m / 2 + C_4 (t_3 + t_4) S / 2 \quad (46)$$

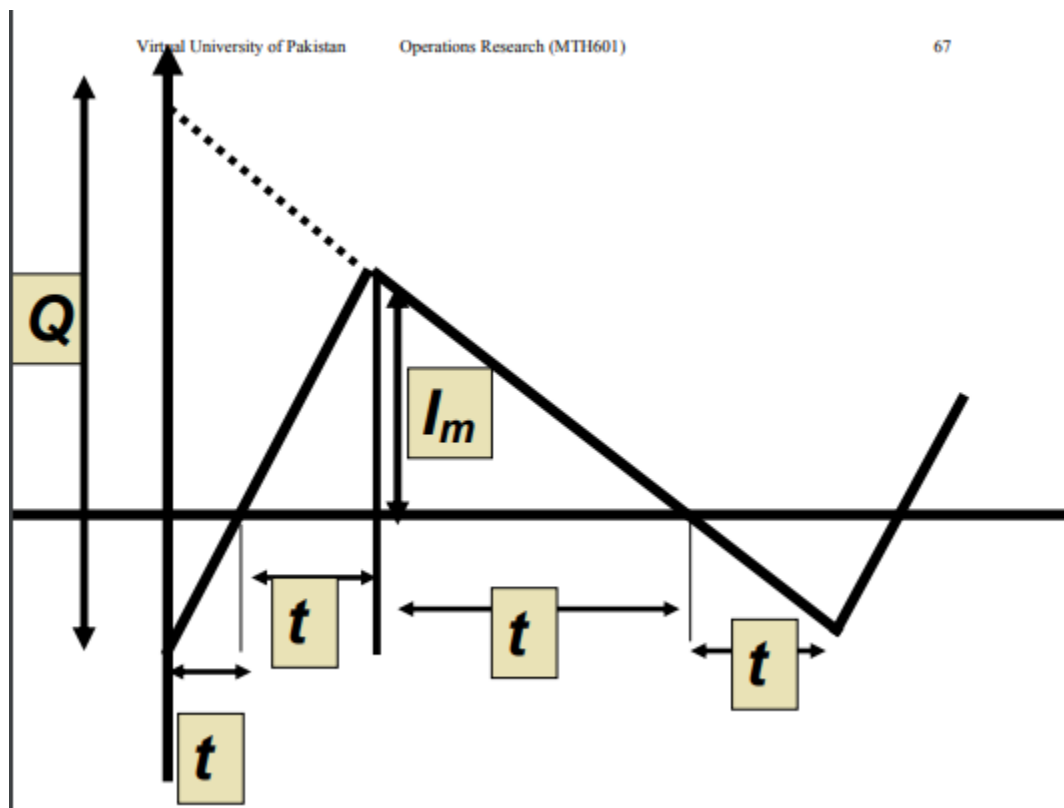
The values of t_1 , t_2 , t_3 and t_4 are to be determined in terms of Q and S , so that the cost is expressed as a function of only two variable Q and S .

Final results are given below without proof.

So
$$Q^* = \sqrt{\frac{2C_2D}{C_3(1-D/R)}} \sqrt{\frac{C_3+C_4}{C_4}} \quad (47)$$

and

$$S^* = \sqrt{\frac{2C_2D}{C_4}} \sqrt{1-D/R} \sqrt{\frac{C_3}{C_3+C_4}}$$



ORDER QUANTITY WITH PRICE-BREAK

The concept of Economic Order Quantity fails in certain cases where there is a discount offered when purchases are made in large quantities. Certain manufacturers offer reduced rate for items when a larger quantity is ordered. It may appear that the inventory holding cost may increase if large quantities of items are ordered. But if the discount offered is so attractive that it even outweighs the holding cost, the probably the order at levels other than the EOQ would be economical. An illustration is given in the following example and the rationale is explained.

Example: A company uses 12000 items per year supplied ordinarily at a price of Rs. 3.00 per item. Carrying costs are 16% of the value of the average inventory and the ordering costs are Rs. 20 per order. The supplier however offers discounts as per the table below:

Order size	Price per item
Less than 2000	Rs. 3.00
2000 to 3999	Rs. 2.90
4000 or more	Rs. 2.85

Compute the economic order size.

$$EOQ = \sqrt{\frac{2 \times 12000 \times 20}{3.00 \times 0.16}} = 1000$$

EOQ at Rs. 2.90 per item

$$= \sqrt{\frac{2 \times 12000 \times 20}{2.90 \times 0.16}} = 1017$$

EOQ at Rs. 2.85 per item

$$= \sqrt{\frac{2 \times 12000 \times 20}{2.85 \times 0.16}} = 1026$$

The EOQ at Rs. 2.90 per item = 1017. But the price per item is Rs. 2.9 only if the items are ordered in the range of 2000 to 3999. This is therefore an infeasible solution. Similarly the EOQ at Rs. 2.85 per item is 1026. This price is valid only for items ordered in the range 4000 or more. This is also an infeasible solution.

We follow the routine procedure and calculate the cost for various order sizes: 1000, 2000, 4000,

	Order size		
	1000	2000	4000
Item cost (Rs.)	36000	34800	34200
Order cost (Rs.)	240	120	60
Holding cost (Rs.)	240	464	912

Exercises

1. Assume the following price structure

Units	Units Price
0-199	Rs. 10.00
200-399	9.75
400-599	9.50
600	9.25

Purchase cost per order = Rs. 25

Cost of the item = Rs. 10

Annual demand = 950 Units

Carrying cost = Rs. 2/Unit/year

2. Find the optimal order quantity for a product for which the price-breaks are as follows:

Items q	Price/Unit
$0 \leq q < 100$	Rs. 20
$100 \leq q < 200$	Rs. 18
$200 \leq q$	Rs. 16

The monthly demand for the product is 600 units. The storage cost is 15% of unit cost and the cost of ordering is Rs. 30 per order.

DYNAMIC ORDER QUANTITY

The basic assumption in the derivation of Economic Order Quantity models discussed previously is that the demand is uniform. But in certain situations the demand is not uniform. It may rise and fall, depending on seasonal influences. A general method is discussed below that can be applied to any pattern of varying demand due to seasonal or irregular variations.

Consider the following example to illustrate how a varying demand problem can be tackled. This is known as **Dynamic Order Quantity model**.

Example: The requirements for 12 months are given below:

Month	1	2	3	4	5	6	7	8	9	10	11	12
Requirement	20	40	10	10	10	2	40	30	40	40	10	20

Set up cost = Rs. 20
 Unit price = Rs. 5 per item.
 Interest = 24% per year
 or 2% per month.

Solution: To calculate the dynamic order quantity we can adopt the following procedure.

The first month's requirement has to be ordered in the first month itself at a procurement cost of Rs. 20. Now we have to decide whether the second month's requirement can also be ordered along with first month's requirement. This involves additional carrying cost, but this will result in saving an extra set up. Hence if the saving on set up costs outweighs the carrying costs, then we include the second month's requirements along with the first month. Similarly a decision can be taken whether to include the third month's requirement in the first month itself. This procedure is followed until the procurement costs and carrying costs are balanced.

Let n represent the month, $n = 1, 2, \dots, 12$. Let R_n be the requirement during n th month and R_{n+1} be the requirement during $(n + 1)$ th month. If the $(n + 1)$ th month requirement namely R_{n+1} is absorbed in n th month itself, the additional procurement cost is saved.

Hence the saving in procurement cost is (C_2/n) . But the additional carrying cost is $C_3 n (R_{n+1})$. If the additional carrying cost is less than the procurement cost, then the $(n+1)$ th month requirement is to be ordered also with n th month.

Hence we have to check whether

$$n R_{n+1} C_3 < C_2/n$$

$$n^2 R_{n+1} < C_2/C_3$$

If the answer to the above inequality is yes, then, the future month's requirements are included in the first month itself.

If the answer is 'no', then that requirement is to be ordered afresh and this is treated as month $n = 1$. In this

example $C_2/C_3 = 20/0.01 = 200$. All the eleven information can be represented in the table as shown.

Month.	Requirement. R_n	n	$n^2 R_{n+1}$	Is $n^2 R_{n+1} < 200$	Action.
1	20	1	40	Yes	include 40 in month 1
2	40	2	40	Yes	include 10 in month 1
3	10	3	90	Yes	include 10 in month 1
4	10	4	160	Yes	include 10 in month 1
5	10	5	50	Yes	include 2 in month 1
6	2	6	1440	No	set up again in month 7
Total	92				
7	40	1	30	Yes	include 10 in 7th month.
8	30	2	160	Yes	include 40 in 7th month.
9	40	3	360	No	set up again in month 10
Total	110				
10	40	1	10	Yes	include 10 in month 10
11	10	2	80	Yes	include 20 in month 10
12	20				
Total	70				

Hence we order three times a year in the first month, seventh month and tenth month, the batch sizes being 92, 110 and 70 respectively.

Exercise: Compute the dynamic EOQ is for the following requirements.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Requirement.	50	100	10	170	150	180	1	260	100	80	150	200

ABC ANALYSIS

The ABC analysis is the analysis that attracts management on those items where the greatest savings can be expected. This is a simple but powerful tool of statistical sampling in the area of inventory control or materials management.

In this analysis, the items are classified or categorized into three classes, A, B and C by their usage value. The usage value is defined as,

The ABC concept is based on Pareto's law that few high usage value items constitute a major part of the capital invested in inventories whereas a large number of items having low usage value constitute an insignificant part of the capital. If too much inventory is kept, the ABC analysis can be performed on a sample. After obtaining the random sample the following steps are carried out for the ABC analysis.

STEP 1: Compute the annual usage value for every item in the sample by multiplying the annual requirements by the cost per unit.

STEP 2: Arrange the items in descending order of the usage value calculated above.

STEP 3: Make a cumulative total of the number of items and the usage value.

STEP 4: Convert the cumulative total of number of items and usage values into a percentage of their grand totals.

STEP 5: Draw a graph connecting cumulative % items and cumulative % usage value. The graph is divided approximately into three segments, where the curve sharply changes its shape. This indicates the three segments A, B and C.

The class A items whose usage values are higher are to be carefully watched and are under the strict and continued scrutiny of the senior inventory control staff. These items should be issued only on indent sanctioned by the staff. The class C items on the other extreme can be placed on the shop floor and the personnel can help themselves without placing a formal requisition. The class B items fall in between A and C.

ABC concept conforms to the consideration implied in the EOQ model. 'A' items have high inventory carrying costs and should therefore be placed with EOQ concept. The 'C' items require very little capital and have therefore low inventory carrying costs. Hence, they can be purchased in bigger lots. 'B' items are usually placed under statistical stock control.

Example: Perform ABC analysis on the following sample of 10 items from an inventory.

Items	1	2	3	4	5	6	7	8	9	10
Annual Usage (Unit)	300	2700	30	1000	50	220	160	800	600	70
Unit Cost (Rs.)	10	15	10	5	4	100	5	5	15	10

Solution:

STEP 1: Calculate usage value by multiplying annual usage of each item with its unit cost and tabulate them and assign rank by giving rank to the largest usage value as in table below.

Items No.	Annual Usage	Ranking
1	3000	6
2	40500	1
3	300	9
4	5000	4
5	200	10
6	22000	2
7	800	7
8	4000	5
9	9000	3
10	700	8

STEP 2: Arrange items as per ranking and calculate cumulative usage and % cumulative value as in the table below.

Rank	Item No.	Annual usage	Cumulative Annual usage	% of cumulative Annual usage	Cumulative % of item
1	2	40500	40500	47	10
2	6	22000	62500	73	20
3	9	9000	71500	84	30
4	4	5000	76500	91	40
5	8	4000	80500	94.2	50
6	1	3000	83500	98	60
7	7	800	84300	98.6	70
8	10	700	85000	99.4	80
9	3	300	85300	99.8	90
10	5	200	85500	100	100

If you draw the figure, you will see that the curve changes at points (say) X and Y. The items upto X is classified as class A items and between X and Y as class B and the rest as class C items.

SOME DEFINITIONS

Lead-time: This is defined as the time interval between the placing of the orders and the actual receipt of goods.

Lead-time Demand: This is the lead-time multiplied by demand rate. For example, if the lead-time is 3 weeks and the demand is at the rate of 50 items per week, then the lead-time demand is $3 \times 50 = 150$ items.

The lead-time may not be constant. For one batch, a vendor may take 45 days and for the next batch 50 days and so on. Lead-time itself is therefore a stochastic variable. This complicates the problem of accumulating stock over the period encompassed by the lead-time. Lead-time may also be forecast exponentially as is done with the demand.

Safety stock or Buffer stock: Lead-time demand is the stock level, which, on the average is sufficient to satisfy the customer's orders as the stocks are being replenished. "On the average" would mean that during this period of replenishment 50% of the customer's order can be filled and the remaining 50% may either be refused or back ordered to be filled later. The reason for this is obvious. Forecast is after all a point estimate only. If the demand is greater than the forecasts, the customers would not be serviced. If the demand is less than the forecasts, overstocking would occur. When these two variables, stock level and service to the customer, are summed up over thousands of stock level and service to the customer, it becomes a major problem for an organisation to find an acceptable compromise between the two. Sometimes the management in an organisation would like to limit the disservice to the customer down to 5% or 10% at the cost of extra stocking. This extra stock in excess of the leadtime demand is called the safety stock. Safety stock may be expressed as percentages of the lead-time demand. It may be computed in different ways.

Reorder level: This is defined as the level of the inventory at which the order is placed. It has generally two components (i) Lead time Demand and (ii) Safety Stock.

Reorder level (ROL) = Lead time demand (LTD) + Safety stock (SS).

Computation of Safety Stock:

As discussed earlier, if the demand exceeds the forecast, the result is bad service to the customer and if the demand is less than the forecast figure, this results in overstocking. Thus there is a forecast error. This error is assumed to be normally distributed, with zero mean. If the standard deviation of the forecast error is calculated, then safety stock may be set with the desired confidence level to result in not more than 5 or 10% shortages etc.

There is another measure of variation, popularly known as mean absolute deviation (MAD), which can be computed far more easily. It can be routinely smoothed every period for obtaining better estimates of the safety stock.

MAD is related to the standard deviation for a normal distribution as given by.

$$MAD = \sqrt{2/\pi} SD.$$

Therefore, $SD = \sqrt{\pi/2} MAD$

Now safety stock = Z (S.D)
 $= Z \sqrt{\pi/2} MAD$

$$= K MAD$$

where K is called service factor $K = Z \sqrt{\pi/2}$

The safety stock is for just one period and has to be extended over the lead time. The necessary formula has been derived by extensive simulation by statisticians and is given below.

$$MADLT = (0.659 + 0.341 LT) MAD$$

Safety stock over the lead-time

$$= K(0.659 + 0.341 LT) MAD$$

$$= Z \sqrt{\pi/2} (0.659 + 0.34 LT) MAD$$

Example: A company uses annually 50000 units of raw materials at a cost Rs. 1.2 per item. Ordering cost of items is Rs. 45 per order and item carrying cost is 15% per year of the average inventory.

1. Find the economic quantity.
2. Suppose that the company follows the EOQ policy and it operates for 300 days a year, that the procurement time is 12 days and find minimum and average inventories.

Solution:

D = Demand = 50000/year No. of days = 300/year

C₁ = Rs. 1.2 per item Lead time = 12 days

C₂ = Rs. 45 per order Safety stock = 500

C₃ = 15% of Rs. 1.2/item/year.

= Rs. 0.18 per item per year.

$$\begin{aligned}
 (1) \quad EOQ &= \sqrt{\frac{2C_2D}{C_3}} \\
 &= \sqrt{\frac{2 \times 45 \times 50000}{0.18}} \\
 &= \sqrt{25000000} \\
 &= 5000
 \end{aligned}$$

$$(2) \quad \text{Requirement per day} = 50000/300$$

$$\text{Lead time demand} = 12 \times 500/3 = 2000$$

$$\text{Safety stock} = 500$$

$$\text{Reorder level} = \text{Lead time demand} + \text{safety stock}$$

$$= 2000 + 500 = 2500$$

$$\text{Maximum inventory} = \text{E.O.Q} + \text{Safety stock}$$

$$= 5000 + 500 = 5500$$

$$\text{Minimum inventory} = \text{Safety stock} = 500$$

$$\text{Average inventory} = (\text{Max. Inv.} + \text{Min. Inv.})/2$$

$$= (5500 + 500)/2 = 3000$$

Example A scrutiny of past records gives the following distributions for lead time and daily demand during lead time.

Lead time distribution

Lead time (days)	3	4	5	6	7	8	9	10
Frequency	2	3	4	4	2	2	2	1

Demand distribution

Demand/day (units)	0	1	2	3	4	5	6	7
Frequency	2	4	5	5	4	2	1	2

What should be the buffer stock?

Solution:

Computation of average or mean lead time.

$$\begin{aligned} &= \frac{\sum (\text{Frequency} \times \text{lead time})}{\sum \text{Frequency}} \\ &= \frac{6+12+20+24+14+6+18+10}{20} \\ &= 6 \text{ days} \end{aligned}$$

Average demand

$$\begin{aligned} &= \frac{\sum (\text{Frequency} \times \text{demand/day})}{\sum \text{frequency}} \\ &= \frac{0+4+10+15+16+10+6+14}{25} \\ &= 3 \end{aligned}$$

Average lead-time demand

$$= \text{Average lead time} \times \text{Average demand/day}$$

$$= 6 \times 3 = 18.$$

Maximum lead-time demand.

$$= \text{Max. lead time} \times \text{Max. demand/day}$$

$$= 10 \times 7 = 70$$

$$\therefore \text{Buffer stock} = \text{Max. lead time demand} - \text{Average lead time demand}$$

***Example 7.7.3**

For a fixed order quantity system find the EOQ, SS, ROL and average inventory for an item with the following data.

Demand = 10000 units

Cost of item = Re. 1

Order cost = Rs. 12

Holding cost = 24%

Post lead times = 13 days

Solution:

$$(1) \quad EOQ = \sqrt{\frac{2C_2D}{C_3}}$$
$$= \sqrt{\frac{2 \times 12 \times 10000}{0.24 \times 1}} = 1000 \text{ items}$$

$$(2) \quad \text{The Average lead time} = (13 + 14 + 15 + 16 + 17)/5$$
$$= 15 \text{ days (30 days time is omitted)}$$

$$\text{Maximum lead time} = 30 \text{ days}$$

$$\text{Safety stock} = (30 - 15) \times 10000 / (30 \times 12) = 420$$

$$(3) \quad \text{Reorder level} = \text{Lead time demand} + \text{Safety stock}$$
$$= 417 + 420 = 837$$

$$(4) \quad \text{Average inventory} = (1420 + 420)/2$$
$$= 1840/2 = 920$$

Example: An airline has determined that 10 spare brake cylinders will give them stock out risk of 30%, whereas 14 will reduce the risk to 15% and 16 to 10%. It takes 3 months to receive items from supplier and the airline has an average of 4 cylinders per month. At what stock level should they reorder assuming that they wish to maintain an 85% service level.

Solution:

$$\text{Lead time demand} = 3 \times 4 = 12 \text{ items}$$

Safety stock at 85% service

$$= 15\% \text{ disservice}$$

or 15% stock out risk

$$= 14 \text{ items}$$

$$\text{Reorder level} = 12 + 14 = 26 \text{ items.}$$

Example: Data on the distribution of lead time for a motor component were collected as shown. Management would like to set safety stock levels that will limit the stock out to 10%.

Lead time (weeks)	1	2	3	4	5	6	7	8
Frequency of occurrence	10	20	70	40	30	10	10	10

How many weeks of safety stock are required to provide the desired service level?

Solution:

Lead time (weeks)	Frequency	Probability	Cumulative Probability
1	10	0.05	0.05
2	20	0.10	0.15
3	70	0.35	0.50
4	40	0.20	0.70
5	30	0.15	0.85
6	10	0.05	0.90
7	10	0.05	0.95
8	10	0.05	1.00
200			

Average lead time = $\sum (\text{lead time} \times \text{frequency}) / \sum \text{frequency}$

$$= (10 + 40 + 210 + 160 + 150 + 60 + 70 + 80) / 200$$

$$= 770 / 200 = 3.85 \text{ weeks}$$

Upto 90% service level max. lead time is 6 weeks.

Hence $6 - 3.85 = 2.15$ weeks of stock would provide the service level of 90%

(Note: If the lead time is given as a continuous time distribution take the mid point.)

Example: Demand for a product during an order period is assumed to be normally distributed with mean of 1000 units and standard deviation of 40 units. What % service can a company expect to provide (i) if it satisfies the average demand only (ii) if it carries a safety stock of 60 units.

Solution:

1. If the company provides only average demand, we can expect only 50% service level.
2. The standard normal variate Z is computed with the following formula.

$$Z = (\text{Safety stock} - 0) / \text{Standard deviation}$$

$$= (60 - 0) / 40 = 1.5$$

The area under normal curve for $Z = 1.5$ is 0.4332.

$\therefore$ Service level = $0.50 + 0.4332 = 0.9332$ or 93.32%

Example: A manufacturer of water filters purchases components in EOQ's of 850 units/order. Total demand averages 12000 components per year and MAD = 32 units per month. If the manufacturer carries a safety stock of 80 units, what service level does this give the firm?

Solution:

$$\text{Standard deviation} = \sqrt{\frac{2}{\pi}} \times \text{MAD} = \text{MAD} / 0.8 = 32 / 0.8 = 40$$

$$Z = (S - S_0) / \text{S.D} = (80 - 0) / 40 = 2$$

The area under normal curve for $Z = 2 = 0.4772$.

Service level = 0.9772 or 97.72%

Example: A firm has normally distributed forecast of usage with MAD = 60 units. It desires a service level, which limits the stock, outs to one order cycle per year.

- (1) How much safety stock should be carried if the order quantity is normally a week's supply?
- (2) How much safety stock should be carried if the order quantity is weeks supply.

Solution:

No. of orders = 52/year.

1 stock out in 52 weeks means a 98 % = (51/52) values

For 98% area, the value of

$Z = 2.05$ (from tables)

$$\text{S.D} = \text{MAD} / 0.8 = 60 / 0.8$$

$$Z = (S - S_0) / \text{S.D} = (S - S_0) / 60 / 0.8$$

$$2.05 = (S - S_0) \times 0.8 / 60$$

$$\text{SS} = 2.05 \times 60 / 0.8 = 154 \text{ units.}$$

(a) Number of orders = 52/5

$$1 \text{ stock out} = (52/5 - 1)$$

$$\text{Service level} = (47/5)/(52/5) = 47/52 = 0.904$$

$$Z = 1.285 \text{ (for area} = 0.404) \text{ MAD} = 60$$

$$1.285 = S S/75$$

$$S S = 75 \times 1.285 = 96 \text{ units.}$$

Segment IV: Linear Programming

Lectures 17- 30

LINEAR PROGRAMMING

Linear programming is a mathematical technique designed to aid managers in allocating scarce resources (such as labor, capital, or energy) among competing activities. It reflects, in the form of a model, the organization's attempt to achieve some objective (frequently, maximizing profit contribution, maximizing rate of return, minimizing costs) in view of limited or constrained resources (available capital or labor, service levels, available machine time, budgets).

The linear programming technique can be said to have a linear objective function that is to be optimized (either maximized or minimized) subject to linear equality or inequality constraints and sign restrictions on the variables. The term linear describes the proportionate relationship of two or more variables. Thus, a given change in one variable will always cause a resulting proportional change in another variable.

Some areas in which linear programming have been applied will be helpful in setting the climate for learning about this important technique.

- (i) A company produces agricultural fertilizers. It is interested in minimizing costs while meeting certain specified levels of nitrogen, phosphate, and potash by blending together a number of raw materials.
- (ii) An investor wants to maximize his or her rate of return by investing in stocks and bonds. The investor can set specific conditions that have to be met including availability of capital.
- (iii) A company wants the best possible advertising exposure among a number of national magazines, and radio and television commercials within its available capital requirements.

- (iv) An oil refinery blends several raw gasoline and additives to meet a car manufacturer's specifications while still maximizing its profits.
- (v) A city wants to maximize the daytime use of recreational properties being proposed for purchase with a limited capital available.

This technique, called linear programming (L.P), is solved in a step-by-step manner called iterations. Each step of the procedure is an attempt to improve on the solution until the "best answer" is obtained or until it is shown that no feasible answer exists.

Formulation of the Linear Programming Problem

To formulate a real-life problem as a linear program is an art in itself. To aid you in this task, it is helpful to isolate the essential elements of the problem as a means of asking what the clients wants and what information can be gained from the data that has been provided.

The first step in formulating a problem is to set forth the objective called the objective function.

A second element of a problem is that there are certain constraints on the company's ability to maximize the total contribution. These constraints are:

- (1) quantity of raw materials available,
- (2) the level of demand for the products, and
- (3) the equipment productive capacity.

A further element that must be considered in the problem is the time period being used. The duration may be either long term or short term. Although time is an important element, it is one that has flexibility so that the time horizon may be changed as long as the restrictions are compatible with the periods under consideration.

The last element is that every product has a likelihood of being made. These products are the dependent or decision variables. Of course, the likelihood of a variable's being in the answer may change with the price or contribution values (usually profit and the nature of the restraints. Yet, at this point there is nothing to indicate that differing chances of occurrence exists for the possibility of making each of the products.

The first stage of solving linear programming problems is to set forth the problem in a mathematical form by defining the variables and the resulting constraints. Generally, the relationship is fairly simple using only elementary algebraic notation. The relationships can be seen by first identifying the decision variables. To aid in using algebraic notation, the decision variables can be represented by symbols such as X, Y, Z.

Next, we must build the objective function. If the goal is to maximize profit, we identify our objective

function as

Maximize total profit or Minimize total loss (cost).

Then we write problem constraints.

These steps are now illustrated by taking some examples.

Example 1: Product Mix

The Regal China Company produces two products daily plates and mugs. The company has limited amounts of two resources used in the production of these products clay and labor. Given these limited resources, the company desires to know how many plates to produce each day, in order to Maximize profit. The two products have the following resource requirements for production and profit per item produced (i.e., the model parameters).

Product	Labor (hours/unit)	Clay (lbs./unit)	Profit (Rs./unit)
Plate	1	4	4
Mug	2	3	5

There are 40 hours of labor and 120 pounds of clay available each day for production.

Formulate this problem as a linear programming model by defining each component of the model separately and then combining the components into a single model.

Decision Variables

The decision confronting management in this problem is how many plates and mugs to produce. As such, there are two decision variables that represent the number of plates and mugs to be produced on a daily basis. The quantities to be produced can be represented symbolically as,

X_1 = the number of plates to produce

X_2 = the number of mugs to produce

The Objective Function

The objective of the company is to Maximize total profit. The company's profit is the sum of the individual profits gained from each plate and mug. As such, profits from plates is determine by multiplying the unit profit for each plate, Rs. 4, by the number of plates produced, X_1 . Likewise, profit derived from mugs is the unit profit of a mug, Rs. 5, multiplied by the number of mugs produced, X_2 . Thus, total profit, Z , can be expressed mathematically as

$$\text{Maximize } Z = 4X_1 + 5X_2$$

where

Z = total profit per day

Rs $4X_1$ = profit from plates

Rs $5X_2$ = profit from mugs

By placing the term Maximize in front of the profit function, the relationship expresses the objective of the firm to Maximize total profit.

Model Constraints

This problem has two resources used for production, which are limited, labor and clay. Production of plates and mugs require both labor and clay. For each plate produce, one hour of labor is required. Therefore, the labor used for the production of plates is $1X_1$ hours. Similarly, each mug requires two hours of labor; the labor used for the production of mugs is $2X_2$ hours. Thus, the labor used by the company is the sum of the individual amounts of labor used for each product.

$$1X_1 + 2X_2$$

However, the amount of labor represented " $1X_1 + 2X_2$ " is limited to 40 hrs per day, thus, the complete labor constraint is

$$1X_1 + 2X_2 \leq 40 \text{ hours}$$

The "less than or equal to ($\leq$)" inequality is employed instead of an equality ($=$) because the forty hours of labor is a maximum limitation that can be used, but not an amount that must be used, but not an amount that must be used. This allows the company more flexibility in that it is not restricted to use the 40 hours exactly, but whatever amount necessary to Maximize profit up to and including forty hours. This means that the possibility of "idle or excess capacity" (i.e., the amount under forty hours not used) exists.

The constraint for pottery clay is formulated in the same way as the labor constraint. Since each plate requires four pounds of clay, the amount of clay used daily for the production of plates is $4X_1$ pounds, and since each mug requires three pounds of clay, the amount of clay used for mugs daily is $3X_2$. Given that amount of clay available for production each day is 120 pounds, the material constraint can be formulated as

$$4X_1 + 3X_2 \leq 120 \text{ pounds}$$

A final restriction is that the number of plates and mugs produced be either zero or a positive value, since it would be impossible to produce negative items. These restrictions are referred to as nonnegative constraints and are expressed mathematically as

$$X_1 \geq 0, X_2 \geq 0$$

The complete linear programming model for this problem can now be summarized as

$$\begin{array}{ll}\text{Maximize} & Z = \text{Rs. } 4X_1 + 5X_2 \\ \text{Subject to} & 1X_1 + 2X_2 \leq 40 \\ & 4X_1 + 3X_2 \leq 120 \\ & X_1, X_2 \geq 0\end{array}$$

The solution of this model will result in numerical values for X_1 and X_2 , which will maximize total profit, Z . As one possible solution, consider $X_1 = 5$ plates and $X_2 = 10$ mugs. First we will substitute this hypothetical solution into each of the constraints in order to make sure that the solution does not require more resources than the constraints show are available.

$$1(5) + 2(10) \leq 40 \qquad 25 \leq 40$$

$$4(5) + 3(10) \leq 120 \qquad 50 \leq 120$$

and

$$4(5) + 3(10) \leq 120 \qquad 50 \leq 120$$

Thus, neither one of the constraints is violated by this hypothetical solution. As such, we say the solution is feasible (i.e., it is possible). Substituting these solution values in the objective function gives $Z = 4(5) + 5(10) = \text{Rs. } 70$. However, the maximum profit.

Now consider a solution of $X_1 = 10$ plates and $X_2 = 20$ mugs, This would result in a profit of

$$Z = \text{Rs. } 4(10) + 5(20) = 40 + 100 = \text{Rs. } 140$$

While this is certainly a better solution in terms of profit, it is also infeasible (i.e., not possible) because it violates the resource constraint or labor:

$$1(10) + 2(20) \leq 40$$

$$50 \leq 40$$

Thus, the solution to this problem must both Maximize profit and not violate the constraints. The actual solution to this model which achieves this objective is $X_1 = 24$ plates and $X_2 = 8$ mugs, with a corresponding profit of Rs. 136.

Example 2 Ingredients Mixing

Fauji Foundation produces a cereal SUNFLOWER, which they advertise as meeting the minimum daily requirements for vitamins A and D. The mixing department of the company uses three main ingredients in making the cereal-wheat, oats, and rice, all three of which contain amounts of vitamin A and D. Given that each box of cereal must contain minimum amounts of vitamin A and D, the company has instructed the mixing department determine how many ounces of each ingredient should go into each box of cereal in order to minimize total cost.

This problem differs from the previous one in that its objective is to minimize cost, rather than Maximize profit.

Each ingredient has the following vitamin contribution and requirement per box.

VITAMIN CONTRIBUTION

Vitamin	Wheat (mg./oz.)	Oats (mg./oz.)	Rice (mg./oz.)	Milligrams Required/Box
A	10	20	08	100
D	07	14	12	70

The cost of one ounce of wheat is Rs. 0.4, the cost of an ounce of oats is Rs. 0.6, and the cost of one ounce of rice is Rs. 0.2.

Decision Variables

This problem contains three decision variables for the number of ounces of each ingredient in a box of cereal:

X_1 = ounces of wheat

X_2 = ounces of oats

X_3 = ounces of rice

The Objective Function

The objective of the mixing department of the Fauji Foundation is to minimize the cost of each box of cereal. The total cost is the sum of the individual costs resulting from each ingredient. Thus, the objective function that is to minimize total cost, Z , is expressed as

$$\text{Minimize } Z = \text{Rs. } 0.4X_1 + 0.6X_2 + 0.2X_3$$

where Z = total cost per box

Rs. $0.4 X_1$ = cost of wheat per box

$0.6 X_2$ = cost of rice per box

$0.2 X_3$ = cost of rice per box

Model Constraints

In this problem the constraints reflect the requirements for vitamin consistency of the cereal. Each ingredient contributes a number of milligrams of the vitamin to the cereal. The constraint for vitamin A is

$$10 X_1 + 20 X_2 + 8 X_3 \geq 100 \text{ milligrams}$$

where $10 X_1$ = vitamin A contribution (in mg.) for wheat

$20 X_2$ = vitamin A contribution (in mg.) for oats

$8X_3$ = vitamin A contribution (in mg.) for rice

Notice that rather than an ($\leq$) inequality, as used in the previous example, this constraint requires a $\geq$ (greater than or minimum requirement specifying that at least 100 mg of vitamin A must be in a box. If a minimum cost solution results so that more than 100 mg is in the cereal mix, which is acceptable, however, the amount cannot be less than 100 mg.

The constraint for vitamin D is constructed like the constraint for vitamin A.

$$7X_1 + 14 X_2 + 12X_3 \geq 70 \text{ milligrams}$$

As in the previous problem there are also nonnegative constraints indicating that negative amounts of each ingredient cannot be in the cereal.

$$X_1, X_2, X_3 \geq 0$$

The L.P. model for this problem can be summarized as

$$\text{Minimize} \quad Z = \text{Rs. } 0.4 X_1 + 0.6 X_2 + 0.2 X_3$$

$$\text{Subject to} \quad 10X_1 + 20 X_2 + 8X_3 \geq 100$$

$$7 X_1 + 14 X_2 + 12 X_3 \geq 70$$

$$X_1, X_2, X_3 \geq 0$$

Example 3 Investment Planning

Mr. Majid Khan has Rs. 70, 000 to investment in several alternatives. The alternative investments are national certificates with an 8.5% return, Defence Savings Certificates with a 10% return, NIT with a 6.5% return, and khas deposit with a return of 13%. Each alternative has the same time until maturity. In addition, each investment alternative has a different perceived risk thus creating a desire to diversify. Majid Khan wants to know how much to invest in each alternative in order to maximize the return.

The following guidelines have been established for diversifying the investments and lessening the risk;

1. No more than 20% of the total investment should be in khas deposit.
2. The amount invested in Defence Savings Certificates should not exceed the amount invested in the other three alternatives.
3. At least 30% of the investment should be in NIT and Defence Savings Certificates.
4. The ration of the amount invested in national certificates to the amount invested in NIT should not exceed one to three.

Decision Variables

There are four decision variables in this model representing the monetary amount invested in each investment alternative.

- X_1 = the amount (Rs.) invested in national certificates
- X_2 = the amount (Rs.) invested in Defence Savings Cert.
- X_3 = the amount (Rs.) invested in NIT.
- X_4 = the amount (Rs.) invested in khas deposit.

The Objective Function

The objective of the investor is to maximize the return from the investment in the four alternatives. The total return is the sum of the individual returns from each separate alternative.

Thus, the objective function is expressed as

$$\text{Maximize} \quad Z = \text{Rs. } .085 X_1 + .100 X_2 + .065 X_3 + .130 X_4$$

Where Z = the total return from all investments

$\text{Rs. } .085 X_1$ = the return from the investment in nat. Cer.

$.100 X_2$ = the return from the investment in certificates of deposit.

$.065 X_3$ = the return from the investment in NIT.

$.130 X_4$ = the return from the investment in khas deposit.

Model Constraints

In this problem the constraints are the guidelines established by the investor for diversifying the total investment. Each guideline will be transformed into a mathematical constraint separately.

Guideline one states that no more than 20% of the total investment should be in khas deposit. Since the total investment will be Rs. 70, 000 (i.e., the investor desires to invest the entire amount), then 20% of Rs. 70, 000 is Rs. 14, 000. Thus, this constraint is

$$X_4 \leq \text{Rs. } 14, 000$$

The second guideline indicates that the amount invested in Defence Savings Cert. should not exceed the

amount invested in the other three alternatives. Since the investment in Defence Savings Cert. is X_2 and the amount

invested in the other alternatives is $X_1 + X_3 + X_4$ the constraint is

$$X_2 \leq X_1 + X_3 + X_4$$

However, the solution technique for linear programming problems will require that constraints be in a

standard form so that all decision variables are on the left side of the inequality (i.e., $\leq$) and all numerical values are on the right side. Thus, by subtracting, $X_1 + X_3 + X_4$ from both sides of the sign, this constraint in proper form becomes

$$X_2 - X_1 - X_3 - X_4 \leq 0$$

Thus third guideline specifies that at least 30% of the investment should be in NIT and Defence Savings Certificates. Given that 30% of the Rs. 70, 000 total is Rs. 21, 000 and the amount invested in Defence Savings Certificates and NIT is represented by $X_2 + X_3$, the constraint is,

$$X_2 + X_3 \geq \text{Rs. } 21,000$$

The fourth guideline states that the ratio of the amount invested in national certificates to the amount invested in NIT should not exceed one to three. This constraint is expressed as

$$(X_1) / (X_3) \leq 1/3$$

This constraint is not in standard linear programming form because of the fractional relationship of the decision variables, X_1/X_3 . It is converted as follows;

$$X_1 \leq 1 X_3/3$$

$$3 X_1 - X_3 \leq 0$$

Finally, Majid Khan wants to invest all of the Rs. 70, 000 in the four alternatives. Thus, the sum of all the investments in the four alternatives must equal Rs. 70, 000,

$$X_1 + X_2 + X_3 + X_4 = \text{Rs. } 70,000$$

This last constraint differs from the $<$ and $>$ inequalities previously developed, in that a specific requirement exists to invest an exact amount. Thus, the possibility of investing more than Rs. 70, 000 or less than Rs. 70, 000 is not considered.

This problem contains all three of the types of constraints that are possible in a linear programming problem: $<$, $=$ and $>$. Further, note that there is no restriction on a model containing any mix of these types of constraints as demonstrated in this problem.

The complete LP model for this problem can be summarized as

Maximize $Z = .085X_1 + .100X_2 + .065X_3 + .130X_4$

Subject to $X_4 \leq 14,000$

$$X_2 - X_1 - X_3 - X_4 \leq 0$$

$$X_2 + X_3 \geq 21,000$$

$$3X_1 - X_3 \leq 0$$

$$X_1 + X_2 + X_3 + X_4 = 70,000$$

$$X_1, X_2, X_3, X_4, \geq 0$$

Example 4 Chemical Mixture

United Chemical Company produces a chemical mixture for a customer in 1,000 - pound batches. The mixture contains three ingredients - zinc, mercury, and potassium. The mixture must conform to formula specifications (i.e., a recipe) supplied by the customer. The company wants to know the amount of each ingredient to put in the mixture that will meet all the requirements of the mix and minimize total cost.

The formula for each batch of the mixture consists of the following specifications:

1. The mixture must contain at least 200 lbs. of mercury.
2. The mixture must contain at least 300 lbs. of zinc.
3. The mixture must contain at least 100 lbs. of potassium.

The cost per pound for mercury is Rs. 4; for zinc, Rs. 8; and for potassium, Rs. 9.

Decision Variables

The model for this problem contains three decision variables representing the amount of each ingredient in the mixture:

X_1 = the number of lbs. of mercury in a batch.

X_2 = the number of lbs. of zinc in a batch.

X_3 = the number of lbs. of potassium in a batch.

The Objective Function

The objective of the company is to minimize the cost of producing a batch of the chemical mixture. The total cost is the sum of the individual costs of each ingredient:

$$\text{Minimize } Z = \text{Rs. } 4X_1 + 8X_2 + 9X_3$$

where

Z = the total cost of all ingredients

Rs. $4X_1$ = the cost of mercury in each batch
 $8X_2$ = the cost of zinc in each batch
 $9X_3$ = the cost of potassium in each batch.

Model Constraints

In this problem the constraints are derived for the chemical formula.

The first specification indicates that the mixture must contain at least 200 lbs. of mercury,

$$X_1 \geq 200$$

The second specification is that the mixture must contain at least 300 lbs. of zinc,

$$X_2 \geq 300$$

The third specification is that the mixture must contain at least 100 lbs. of potassium,

$$X_3 \geq 100$$

Finally, it must not be overlooked that the whole mixture relates to a 1,000-lb. batch. As such, the sum of

all ingredients must exactly equal 1,000 lbs.,

$$X_1 + X_2 + X_3 = 1,000$$

The complete linear programming model can be summarized as

$$\text{Minimize } Z = 4X_1 + 8X_2 + 9X_3$$

Subject to $X_1 \geq 200$

$$X_2 \geq 300$$

$$X_3 \geq 100$$

$$X_1 + X_2 + X_3 = 1,000$$

$$X_1, X_2, X_3 \geq 0$$

Example 5 Marketing

The Bata Shoe Company has contracted with an advertising firm to determine the types and amount of advertising it should have for its stores. The three types of advertising available are radio and television commercials and newspaper ads. The retail store desires to know the number of each type of advertisement it should purchase in order to Maximize exposure. It is estimated that each ad and commercial will reach the following potential audience and cost the following amount.

Type of Advertisement	Exposure (people/ad or commercial)	Cost
Television commercial	20,000	Rs. 15,000
Radio commercial	12,000	8,000
Newspaper ad	9,000	4,000

The following resource constraints exist:

1. There is a budget limit of Rs. 100,000 available for advertising.
2. The television station has enough time available for four commercials.
3. The radio station has enough time available for ten radio commercials.
4. The newspaper has enough space available for seven ads.
5. The advertising agency has time and staff to produce at most a total of fifteen commercials ads.

Decision Variables

This model consists of three decision variables representing the number of each type of advertising produced:

X_1 = the number of television commercials

X_2 = the number of radio commercials

X_3 = the number of newspaper ads

The Objective Function

The objective of this problem is different from the objectives in the previous examples in which only profit was Maximized (or cost minimized). In this problem profit is not Maximized, but rather the audience exposure is Maximized.

This objective function demonstrates that although a linear programming model must either Maximize or Minimize some objective, the objective itself can be in terms of any type of activity or valuation.

For this problem the objective of audience exposure is determined by summing the audience exposure gained from each type of advertising

$$\text{Maximize} \quad Z = 20,000 X_1 + 12,000 X_2 + 9,000 X_3$$

Where Z = the total number of audience exposures

$20,000 X_1$ = the estimated number of exposures from television commercials

$12,000 X_2$ = the estimated number of exposures from radio commercials

$9,000 X_3$ = the estimated number of exposures from newspaper ads

Model Constraints

The first constraint in this model reflects the limited budget of Rs. 100,000 allocated for advertisement,

$$\text{Rs.} \quad 15,000 X_1 + 6,000 X_2 + 4,000 X_3 < 100,000$$

where

Rs. $15,000 X_1$ = the amount spent for television advertising

$6,000 X_2$ = the amount spent for radio advertising

$4,000 X_3$ = the amount spent for newspaper advertising

The next three constraints represent the fact that television and radio commercials are limited to four and ten, respectively, while newspaper ads are limited to seven.

$$X_1 \leq 4 \text{ commercials}$$

$$X_2 \leq 10 \text{ commercials}$$

$$X_3 \leq 7 \text{ ads}$$

The final constraint specifies that the total number of commercials and ads cannot exceed fifteen due to the limitations of the advertising firm:

$$X_1 + X_2 + X_3 < 15 \text{ commercials and ads}$$

The complete linear programming model for this problem is summarized as

$$\text{Maximize} \quad Z = 20,000 X_1 + 12,000 X_2 + 9,000 X_3$$

Subject to

$$\text{Rs. } 15,000 X_1 + 6,000 X_2 + 4,000 X_3 < \text{Rs. } 100,000$$

$$X_1 \leq 4$$

$$X_2 \leq 10$$

$$X_3 \leq 7$$

$$X_1 + X_2 + X_3 \leq 15$$

$$X_1, X_2, X_3 > 0$$

Example 6 Transportation

The Philips Television Company produces and ships televisions from three warehouses to three retail stores on a monthly basis. Each warehouse has a fixed demand per month. The manufacturer wants to know the number of television sets to ship from each warehouse to each store in order to minimize the total cost of transportation.

Each warehouse has the following supply of televisions available for shipment each month.

Warehouse	Supply (sets)
1. Karachi	300
2. Lahore	100
3. Islamabad	200

	600

Each retail store has the following monthly demand for television sets:

Store	Demand (sets)
A. Faisalabad	150
B. Peshawar	250
C. Hyderabad	200

	600

The costs for transporting television sets from each warehouse to each retail store are different as a result of different modes of transportation and distances. The shipping cost per television set for each route are,

From Warehouse	To store		
	A	B	C
1	Rs. 6	Rs. 8	Rs. 1
2	4	2	3
3	3	5	7

Decision Variables

The model for this problem consists of nine decision variables representing the number of television sets transported from each of the three warehouses to each of the three stores,

X_{ij} = the No. of television sets shipped from warehouse "i" to store "j" where $i = 1, 2, 3$ and $j = A, B, C$.

X_{ij} is referred to as a double subscripted variable. However, the subscript, whether double or single simply gives a "name" to the variable (i.e., distinguishes it from other decision variables). As such, the reader should not view it as more complex than it actually is. For example, the decision variable X_{3A} Islamabad to store A in Faisalabad.

The Objective Function

The objective function of the television manufacturer is to minimize the total transportation costs for all shipments. Thus, the objective function is the sum of the individual shipping costs from each warehouse to each store.

$$\text{Minimize } Z = \text{Rs. } 6X_{1A} + 8X_{1B} + 1X_{1C} + 4X_{2A} + 2X_{2B} + 3X_{2C} + 3X_{3A} + 5X_{3B} + 7X_{3C}$$

Model Constraints

The constraints in this model are available television sets at each warehouse and the number of sets demanded at each store. As such, six constraints exist -- one for each warehouse's supply and one for each store's demand. For example, warehouse 1 retail stores. Since the amount shipped to the three stores is the sum of X_{1A} , X_{1B} , and X_{1C} the constraint for warehouse 1 is

$$X_{1A} + X_{1B} + X_{1C} = 300$$

This constrain is an equality (=) for two reasons. First, more than 300 television sets cannot be shipped, because that is cannot be shipped, because all 300 are needed at the three stores, the three warehouses must supply all that can be supplied. Thus, since the total shipped from warehouse 1 cannot exceed 300 or be less than 300 the constraint is equality. Similarly, the other two supply constraints for warehouse 2 and 3 are also equalities.

$$X_{2A} + X_{2B} + X_{2C} = 100$$

$$X_{3A} + X_{3B} + X_{3C} = 200$$

The three demand constraints are developed in the same way except that television sets can be supplied from any of the three warehouses. Thus, the amount shipped to one store is the sum of the shipments from the three warehouses:

$$X_{1A} + X_{2A} + X_{3A} = 150$$

$$X_{1B} + X_{2B} + X_{3B} = 250$$

$$X_{1C} + X_{2C} + X_{3C} = 200$$

The complete linear programming model for this problem is summarized as:

Minimize $Z = \text{Rs. } 6X_{1A} + 8X_{1B} + 1X_{1C} + 4X_{2A} + 2X_{2B} + 3X_{2C} + 3X_{3A} + 5X_{3B} + 7X_{3C}$

subject to

$$X_{1A} + X_{1B} + X_{1C} = 300$$

$$X_{2A} + X_{2B} + X_{2C} = 100$$

$$X_{3A} + X_{3B} + X_{3C} = 200$$

$$X_{1A} + X_{2A} + X_{3A} = 150$$

$$X_{1B} + X_{2B} + X_{3B} = 250$$

$$X_{1C} + X_{2C} + X_{3C} = 200$$

$$X_{ij} \geq 0$$

LINEAR PROGRAMMING PROBLEM

Let x_i = decision variable for i th variable.

c_i = profit or cost co-efficient of i th variable.

Z = function to be maximized or minimized.

Thus for n decision variables, the objective function is to maximize or minimize

$$Z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

Let a_{ij} = co-efficient of the j th constraint and i th variable

b_i = resource limitation for i th constraint

Thus the restrictions may be expressed in the general form

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m$$

and $x_i \geq 0$ for all values of i from 1 to n

The same linear programming problem can be expressed in a more condensed form using summation notation or matrix equation.

$$\text{Maximize } Z = \sum_{i=1}^n c_i x_i$$

$$\text{Subject to } \sum_{j=1}^n a_{ij} x_j \text{ for all } i = 1, 2, \dots, m$$

$$\text{and } x_j \geq 0 \text{ for all } j = 1, 2, \dots, n$$

or

$$\text{Maximize } Z = CX, \text{ subject to } AX \leq B$$

where C is a row vector and X and B are column vectors. A is a co-efficient matrix of the order $m \times n$.

The linear programming problem may have an objective function to minimize cost also. The inequalities may be "greater than or equal" instead of "less than or equal". Further in some cases, the restrictions involve "equalities".

The successful application of linear programming is the ability to recognize that the problem can be formulated as a linear programming model.

REVIEW QUESTIONS

1. What do you understand by Linear Programming problem?
2. Explain how linear programming can be applied to management problems.
3. Explain the terms: objective function and restrictions in relation to linear programming problem.
4. Give a mathematical format in which a linear programming problem is expressed.
5. Enumerate the limitations of linear programming problem.
6. In relation to linear programming explain the implications of the following assumptions of the model.
 - Linearity for the objective function and constraints.
 - Continuous variables.
 - Certainty.
7. Discuss in brief linear programming as a technique for resource utilization.

8. A company makes products A, B, C and D which flow through four departments: Drilling, Milling, Lathe and Assembly. The variable time per unit of different products are given below in hours:

Product	Drilling	Milling	Lathe	Assembly
A	3	0	3	4
B	7	2	4	6
C	4	4	0	5
D	0	6	5	3

The unit contribution of the four products and hours of availability in the four departments are:

Product	Contribution/Unit Rs.	Department	Hours Available
A	9	Drilling	70
B	18	Milling	80
C	14	Lathe	90
D	11	Assembly	100

Formulate a linear programme for maximizing the contribution.

9. A pension fund manager is considering investing in two shares A and B. It is estimated that:
- Share A will earn a dividend of 12% per annum and share B 4 % per annum.
 - Growth in the market value in one year of share will be 10 paise per Re. 1 invested and in B 40 paise per Re. 1 invested.

He requires investing the minimum total sum, which will give

- dividend income of at least Rs. 600 per annum and
 - growth in one year of atleast Rs. 1000 on the initial investment. you are required to state the mathematical formulation of the problem.
10. A manufacturer uses three raw products a, b, c priced at 30, 50, 120 rupees per kg respectively. He can make three different products A, B and C, which can be sold at 90, 100, 120 rupees per kg respectively. The raw products can be obtained only in limited quantities, namely 20, 15 and 10 kg per day. Given: 2 kg of a plus 1 kg of b plus 1 kg of c will yield 4 kg of A; 3 kg of a plus 2 kg of b plus 2kg of c will yield 7 kg of B; 2kg of b plus 1 kg of c will yield 3 kg of C.

Make a production plan, assuming that the other costs are not influenced by the choice among the alternatives. Formulate the model of the problem.

11. A marketing manager wishes to allocate his annual advertising budget of Rs. 20,000 in two media vehicles A and B. The unit of a message in media A is Rs. 1000 and that of B is Rs. 1500, Media A is a monthly magazine and not more than one insertion is desired in one issue. At least 5 messages should appear in media B. The expected effective audience for unit messages in the media A is 40, 000 and media B is 55, 000.

- Develop a mathematical model.
- Solve for maximizing the total effective audience.

12. A company produces four products 1 to 4. Raw material requirements, storage space needed, production rates and profits are given in the table below. The total amount of raw material available per day for all four products is 180 kg, total space available for storage is 230 sq. mtr. and 7 hours/day is used for production.

	1	2	3	4
Raw materials (Kg/piece)	2	2	1.5	4
Space (sq. mtr./piece)	2	2.5	2	1.5
Production rate (pieces/hr)	15	30	10	15
Profit (Rs./piece)	55	56.6	55	55.5

How many units of each product should be produced to maximize total profit?

13. A ship has three cargo holds, forward, aft and centre.

The capacity limits are:

Forward	2000 tons	1000 cubic meter
Centre	3000 tons	1350 cubic meter
Aft	1500 tons	300 cubic meter

The following cargoes are offered; the ship owners may accept all or any part of each commodity.

Commodity	Amount (tons)	Volume/ton (cubic meter)	Profit/ton (Rs.)
A	6000	60	600
B	4000	50	800
C	2000	25	500

In order to preserve the trim of the ship, the weight in each hold must be proportional to the capacity in tons. How should cargo be distributed so as to maximize profit? (only formulation of the problem needed). Page 101